

Then

$$\begin{aligned}
 \gamma_n &= \sum_{r=n}^{\infty} \delta_r U_{r-n} V_{r+n} \\
 &= \sum_{r=n}^{\infty} \frac{(\rho_1; q)_r (\rho_2; q)_r (q^{-N}; q)_r q^r}{(\rho_1 \rho_2 q^{-N}/a; q)_r (q; q)_{r-n} (aq; q)_{r+n}} \\
 &= \sum_{r=0}^{\infty} \frac{(\rho_1; q)_{r+n} (\rho_2; q)_{r+n} (q^{-N}; q)_{r+n} q^{r+n}}{(\rho_1 \rho_2 q^{-N}/a; q)_{r+n} (q; q)_r (aq; q)_{r+2n}} \\
 &= \frac{(\rho_1; q)_n (\rho_2; q)_n (q^{-N}; q)_n q^n}{(\rho_1 \rho_2 q^{-N}/a; q)_n (aq; q)_{2n}} {}_3\phi_2 \left(\begin{matrix} \rho_1 q^n, \rho_2 q^n, q^{-(N-n)} \\ \rho_1 \rho_2 q^{n-N}/a, aq^{2n+1} \end{matrix}; q, q \right) \\
 &= \frac{(\rho_1; q)_n (\rho_2; q)_n (q^{-N}; q)_n q^n (aq^{n+1}/\rho_1; q)_{N-n} (aq^{n+1}/\rho_2; q)_{N-n}}{(\rho_1 \rho_2 q^{-N}/a; q)_n (aq; q)_{2n} (aq^{2n+1}; q)_{N-n} (aq/\rho_1 \rho_2; q)_{N-n}} \\
 &= \frac{(aq/\rho_1; q)_N (aq/\rho_2; q)_N}{(aq; q)_N (aq/\rho_1 \rho_2; q)_N} \frac{(-1)^n (\rho_1; q)_n (\rho_2; q)_n (q^{-N}; q)_n}{(aq/\rho_1; q)_n (aq/\rho_2; q)_n (aq^{N+1}; q)_n} \\
 &\quad \times (aq/\rho_1 \rho_2)^n q^{nN-n(n-1)/2}.
 \end{aligned}$$

We now turn to the proof of (12.2.3). Note that

$$\begin{aligned}
 &\sum_{r=0}^N \frac{\alpha'_r}{(q; q)_{N-r} (aq; q)_{N+r}} \\
 &= \sum_{r=0}^N \frac{(\rho_1; q)_r (\rho_2; q)_r (aq/\rho_1 \rho_2)^r \alpha_r}{(aq/\rho_1; q)_r (aq/\rho_2; q)_r (q; q)_{N-r} (aq; q)_{N+r}} \\
 &= \sum_{r=0}^N \frac{(-1)^r (\rho_1; q)_r (\rho_2; q)_r (q^{-N}; q)_r}{(aq/\rho_1; q)_r (aq/\rho_2; q)_r (aq; q)_{N+r}} (aq/\rho_1 \rho_2)^r \cdot q^{rN-r(r-1)/2} \alpha_r \\
 &= \frac{(aq/\rho_1 \rho_2; q)_N}{(aq/\rho_1; q)_N (aq/\rho_2; q)_N} \sum_{r=0}^N \gamma_r \alpha_r \\
 &= \frac{(aq/\rho_1 \rho_2; q)_N}{(aq/\rho_1; q)_N (aq/\rho_2; q)_N} \sum_{r=0}^N \beta_r \delta_r \quad (\text{by Lemma 12.2.2}) \\
 &= \frac{(aq/\rho_1 \rho_2; q)_N}{(aq/\rho_1; q)_N (aq/\rho_2; q)_N} \sum_{r=0}^N \frac{(\rho_1; q)_r (\rho_2; q)_r (q^{-N}; q)_r q^r \beta_r}{(\rho_1 \rho_2 q^{-N}/a; q)_r} \\
 &= \beta'_N,
 \end{aligned}$$

where the last equation follows after an algebraic simplification that produces the expression given in (12.2.4). The theorem is proved. ■

Remark 12.2.1 The Weak Bailey Lemma (Lemma 12.2.1) follows from the full Bailey Lemma by letting $n, \rho_1, \rho_2 \rightarrow \infty$.

We call (α_n, β_n) a Bailey pair if they are related as in Bailey's lemma. The power of the full Bailey Lemma is that, given a Bailey pair (α_n, β_n) a new Bailey pair is produced. Consequently, from one such pair one can construct an infinite sequence $(\alpha_n, \beta_n) \rightarrow (\alpha'_n, \beta'_n) \rightarrow (\alpha''_n, \beta''_n) \rightarrow \dots$ of Bailey pairs by successive application of Bailey's Lemma. This sequence is called a Bailey chain. The simplest conceivable chain starts with

$$\beta_n = \delta_{n,0} = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{if } n > 0. \end{cases} \quad (12.2.5)$$

It is a simple exercise to show that the corresponding α_n is

$$\alpha_n = \frac{(1 - aq^{2n})(a; q)_n (-1)^n q^{\binom{n}{2}}}{(1 - a)(q; q)_n}. \quad (12.2.6)$$

Remark 12.2.2 The fact that (α_n, β_n) is a Bailey pair follows immediately from a formula of Agarwal [1953],

$$\sum_{j=0}^M \frac{(1 - aq^{2j})(q^{-n}; q)_j (a; q)_j q^{nj}}{(1 - a)(aq^{n+1}; q)_j (q; q)_j} = \frac{(aq; q)_M q^{nM} (q^{1-n}; q)_M}{(q; q)_M (aq^{n+1}; q)_M}, \quad (12.2.7)$$

a result that can be proved directly by induction on M . Another way is to use the “inversion” formula: If β_n is given by (12.2.1), then

$$\alpha_n = (1 - aq^{2n}) \sum_{j=0}^n \frac{(aq; q)_{n+j-1} (-1)^{n-j} q^{\binom{n-j}{2}} \beta_j}{(q; q)_{n-j}}. \quad (12.2.8)$$

12.3 Watson's Transformation Formula

In this section we use the Bailey chain to derive some important formulas. In particular, we obtain a q -analog of Whipple's transformation due to Watson [1929]. This transforms a terminating very well poised ${}_8\phi_7$ to a terminating balanced ${}_4\phi_3$. The phrase “very well poised” is defined below.

We start with a simpler result. Recall that (α_n, β_n) in (12.2.5) and (12.2.6) form a Bailey pair. By Theorem 12.2.3, the next pair in the chain, namely (α'_n, β'_n) , is given by

$$\begin{aligned} \beta'_n &= \frac{(aq/\rho_1\rho_2; q)_n}{(q; q)_n (aq/\rho_1; q)_n (aq/\rho_2; q)_n}, \\ \alpha'_n &= \frac{(\rho_1; q)_n (\rho_2; q)_n (aq/\rho_1\rho_2)^n (1 - aq^{2n})(a; q)_n (-1)^n q^{\binom{n}{2}}}{(aq/\rho_1; q)_n (aq/\rho_2; q)_n (1 - a)(q; q)_n}. \end{aligned}$$

And the relation

$$\beta'_n = \sum_{r=0}^n \frac{\alpha'_r}{(q; q)_{n-r} (aq; q)_{n+r}},$$

when written out in full, is

$$\begin{aligned} & \frac{(aq/\rho_1\rho_2)_n}{(q)_n(aq/\rho_1)_n(aq/\rho_2)_n} \\ &= \sum_{r=0}^n \frac{(\rho_1)_r(\rho_2)_r(aq/\rho_1\rho_2)^r(1-aq^{2r})(a)_r(-1)^r q^{\binom{r}{2}}}{(q)_{n-r}(aq)_{n+r}(aq/\rho_1)_r(aq/\rho_2)_r(1-a)(q)_r}, \end{aligned} \quad (12.3.1)$$

where $(x)_r = (x; q)_r$. Note that

$$\begin{aligned} \frac{1-aq^{2r}}{1-a} &= \frac{(1-\sqrt{a}q^r)(1+\sqrt{a}q^r)}{(1-\sqrt{a})(1+\sqrt{a})} = \frac{(\sqrt{a}q; q)_r(-\sqrt{a}q; q)_r}{(\sqrt{a}, q)_r(-\sqrt{a}; q)_r}, \\ (q)_{n-r} &= \frac{(-1)^r (q)_n}{(q^n)_r q^{r(2n-r+1)/2}}, \end{aligned}$$

and

$$(aq)_{n+r} = (aq)_n (aq^{n+1})_r.$$

Thus (12.3.1) is equivalent to the formula

$${}_6\phi_5 \left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, \rho_1, \rho_2, q^{-n} \\ \sqrt{a}, -\sqrt{a}, aq/\rho_1, aq/\rho_2, aq^{n+1} \end{matrix}; q, \frac{aq^{n+1}}{\rho_1\rho_2} \right) = \frac{(aq)_n(aq/\rho_1\rho_2)_n}{(aq/\rho_1)_n(aq/\rho_2)_n}. \quad (12.3.2)$$

The ${}_6\phi_5$ is well poised because the product of a numerator parameter with the corresponding denominator parameter is aq . Additionally, the presence of the factor $(1-aq^{2r})/(1-a)$ now makes it a very well poised ${}_6\phi_5$.

Watson's transformation is obtained by moving up the Bailey chain to (α'', β'') . The relation

$$\sum_{r=0}^n \frac{\alpha''_r}{(q)_{n-r} (aq)_{n+r}} = \beta''_n$$

is seen to be equivalent to

$$\begin{aligned} & \sum_{r=0}^n \frac{(\lambda_1)_r(\lambda_2)_r(aq/\lambda_1\lambda_2)^r(\rho_1)_r(\rho_2)_r(aq/\rho_1\rho_2)^r(1-aq^{2r})(a)_r(-1)^r q^{\binom{r}{2}}}{(aq/\lambda_1)_r(aq/\lambda_2)_r(aq/\rho_1)_r(aq/\rho_2)_r(1-a)(q)_r(q)_{n-r}(aq)_{n+r}} \\ &= \sum_{j \geq 0} \frac{(\lambda_1)_j(\lambda_2)_j(aq/\lambda_1\lambda_2)_{n-j}(aq/\lambda_1\lambda_2)^j(aq/\rho_1\rho_2)_j}{(q)_{n-j}(aq/\lambda_1)_j(aq/\lambda_2)_j(q)_j(aq/\rho_1)_j(aq/\rho_2)_j}. \end{aligned} \quad (12.3.3)$$

By the formulas used to obtain (12.3.2), it is easy to show that (12.3.3) can be written as follows:

$$\begin{aligned} & {}_8\phi_7 \left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, \lambda_1, \lambda_2, \rho_1, \rho_2, q^{-n} \\ \sqrt{a}, -\sqrt{a}, aq/\lambda_1, aq/\lambda_2, aq/\rho_1, aq/\rho_2, aq^{n+1} \end{matrix}; q, \frac{a^2 q^{n+2}}{\lambda_1 \lambda_2 \rho_1 \rho_2} \right) \\ &= \frac{(aq)_n (aq/\rho_1 \rho_2)_n}{(aq/\rho_1)_n (aq/\rho_2)_n} {}_4\phi_3 \left(\begin{matrix} aq/\lambda_1 \lambda_2, \rho_1, \rho_2, q^{-n} \\ aq/\lambda_1, aq/\lambda_2, \rho_1 \rho_2 q^{-n}/a \end{matrix}; q, q \right). \end{aligned} \quad (12.3.4)$$

This is Watson's formula. When $aq/\lambda_1 \lambda_2 = \rho_1 \rho_2 q^{-n}/a$, the ${}_4\phi_3$ becomes a balanced ${}_3\phi_2$, which can be summed by the q -Pfaff–Saalschütz formula (10.11.3). The result in this case is the q -analog of Dougall's formula due to Jackson [1921]:

$$\begin{aligned} & {}_8\phi_7 \left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, \lambda_1, \lambda_2, \rho_1, \rho_2, q^{-n} \\ \sqrt{a}, -\sqrt{a}, aq/\lambda_1, aq/\lambda_2, aq/\rho_1, aq/\rho_2, aq^{n+1} \end{matrix}; q, q \right) \\ &= \frac{(aq)_n (aq/\rho_1 \rho_2)_n (aq/\lambda_1 \rho_1)_n (aq/\lambda_1 \lambda_2)_n}{(aq/\lambda_1)_n (aq/\lambda_2)_n (aq/\rho_1)_n (aq/\lambda_1 \lambda_2 \rho_1)_n}, \quad \text{when } a^2 q = \lambda_1 \lambda_2 \rho_1 \rho_2 q^{-n}. \end{aligned} \quad (12.3.5)$$

Let $n \rightarrow \infty$ in (12.3.5). This gives

$$\begin{aligned} & {}_6\phi_5 \left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, \lambda_1, \lambda_2, \rho_1 \\ \sqrt{a}, -\sqrt{a}, aq/\lambda_1, aq/\lambda_2, aq/\rho_1 \end{matrix}; q, \frac{aq}{\lambda_1 \lambda_2 \rho_1} \right) \\ &= \frac{(aq)_\infty (aq/\rho_1 \rho_2)_\infty (aq/\lambda_1 \rho_1)_\infty (aq/\lambda_1 \lambda_2)_\infty}{(aq/\lambda_1)_\infty (aq/\lambda_2)_\infty (aq/\rho_1)_\infty (aq/\lambda_1 \lambda_2 \rho_1)_\infty}. \end{aligned} \quad (12.3.6)$$

This is a nonterminating form of (12.3.2).

Now observe that the ${}_8\phi_7$ in (12.3.4) is symmetric in $\lambda_1, \lambda_2, \rho_1, \rho_2$ and hence

$$\begin{aligned} & \frac{(aq/\rho_1 \rho_2)_n}{(aq/\rho_1)_n (aq/\rho_2)_n} {}_4\phi_3 \left(\begin{matrix} aq/\lambda_1 \lambda_2, \rho_1, \rho_2, q^{-n} \\ aq/\lambda_1, aq/\lambda_2, \rho_1 \rho_2 q^{-n}/a \end{matrix}; q, q \right) \\ &= \frac{(aq/\lambda_1 \lambda_2)_n}{(aq/\lambda_1)_n (aq/\lambda_2)_n} {}_4\phi_3 \left(\begin{matrix} aq/\rho_1 \rho_2, \lambda_1, \lambda_2, q^{-n} \\ aq/\rho_1, aq/\rho_2, \lambda_1 \lambda_2 q^{-n}/a \end{matrix}; q, q \right). \end{aligned} \quad (12.3.7)$$

A q -analog of Dixon's formula for a well-poised ${}_3F_2$ is obtained by setting $\rho_1 = \sqrt{a}$ in (12.3.6):

$$\begin{aligned} & {}_4\phi_3 \left(\begin{matrix} a, -q\sqrt{a}, \lambda_1, \lambda_2 \\ -\sqrt{a}, aq/\lambda_1, aq/\lambda_2 \end{matrix}; q, \frac{q\sqrt{a}}{\lambda_1 \lambda_2} \right) \\ &= \frac{(aq)_\infty (aq/\lambda_1 \lambda_2)_\infty (q\sqrt{a}/\lambda_1)_\infty (q\sqrt{a}/\lambda_2)_\infty}{(aq/\lambda_1)_\infty (aq/\lambda_2)_\infty (q\sqrt{a})_\infty (q\sqrt{a}/\lambda_1 \lambda_2)_\infty}. \end{aligned}$$

The two Rogers–Ramanujan formulas can also be derived from Watson’s transformation (12.3.4). Let $\lambda_1, \lambda_2, \rho_1, \rho_2 \rightarrow \infty$ to get

$$\begin{aligned} & \sum_{j=0}^n \frac{(a; q)_j (1 - aq^{2j})(q^{-n}; q)_j (a^2 q^{n+2j})^j}{(q; q)_j (1 - a)(aq^{n+1}; q)_j} \\ &= (aq; q)_n \sum_{j=0}^n \frac{(-1)^j (q^{-n}; q)_j}{(q; q)_j} (aq^{n+1+(j-1)/2})^j. \end{aligned}$$

Now let $n \rightarrow \infty$ and apply the dominated convergence theorem. After simplification, the result is

$$1 + \sum_{j=1}^{\infty} \frac{(aq; q)_{j-1} (1 - aq^{2j})}{(q; q)_j} (-1)^j a^{2j} q^{j(5j-1)/2} = (aq; q)_{\infty} \sum_{j=0}^{\infty} \frac{a^j q^{j^2}}{(q; q)_j}.$$

For $a = 1$, this is the first Rogers–Ramanujan identity and for $a = q$ it is the second one.

If one wants to generalize a q -series identity that has quadratic powers in it, it is natural to try to replace each $q^{\binom{k}{2}}$ with $(a; q)_k$ times something else that will give $q^{\binom{k}{2}}$ using

$$\lim_{a \rightarrow \infty} (a; q)_k / (-a)^k = q^{\binom{k}{2}}.$$

Since there are two Rogers–Ramanujan identities and five factors $q^{\binom{k}{2}}$, the minimum extension of the Rogers–Ramanujan identities of this type should have six free parameters. Formula (12.3.4) is this extension.

12.4 Other Applications

The summation formula for a very well poised ${}_6\phi_5$ and Watson’s transformation were fairly straightforward consequences of Bailey’s Lemma. Further interesting results are possible by starting with the Bailey pair (12.2.5) and (12.2.6) and moving further along the chain. For example,

$$\begin{aligned} & \sum_{n_1 \geq n_2 \geq \dots \geq n_k \geq 0} \frac{q^{n_1^2 + n_2^2 + \dots + n_k^2}}{(q)_{n_1 - n_2} (q)_{n_2 - n_3} \cdots (q)_{n_{k-1} - n_k} (q)_{n_k}} \\ &= \prod_{\substack{n=1 \\ n \not\equiv 0, \pm(k+1) \pmod{2k+3}}}^{\infty} \frac{1}{1 - q^n}. \end{aligned}$$

This identity reduces to the first Rogers–Ramanujan identity when $k = 1$. As noted before, it can be proved by examining $(\alpha_n^{(k+1)}, \beta_n^{(k+1)})$ in the Bailey chain starting with the pair (12.2.5) and (12.2.6).

It should be stressed that this is not the only Bailey chain of interest. In a totally different context, another of Ramanujan's series,

$$S(q) := \sum_{n=0}^{\infty} \frac{q^{\binom{n+1}{2}}}{(-q; q)_n} =: \sum_{n=0}^{\infty} s_n q^n,$$

was studied. The relevant Bailey pair is

$$\beta_n = \frac{(-q)^n}{(q^2; q^2)_n}$$

and

$$\alpha_n = q^{n^2+n} \sum_{j=-n}^n (-1)^j q^{-j^2}.$$

From

$$\beta'_n = \sum_{r=0}^n \frac{\alpha'_r}{(q; q)_{n-r} (q; q)_{n+r}},$$

we can deduce

$$S(q) = \sum_{\substack{n \geq 0 \\ |j| \leq n}} (-1)^{n+j} q^{n(3n+1)/2 - j^2} (1 - q^{2n+1}).$$

From this result it can be shown that almost all s_n equal zero and that for any integer M (positive, negative, or zero), there exist infinitely many n so that $s_n = M$. For details, see Andrews, Dyson, and Hickerson [1988]. This paper also contains references to the work of Slater and Bressoud in Exercises 3, 4, and 5.

Exercises

1. Obtain the second Rogers–Ramanujan identity (12.1.2) from the weak Bailey Lemma 12.2.1.
2. Prove Agarwal's formula (12.2.7), namely

$$\sum_{j=0}^M \frac{(1 - aq^{2j})(q^{-n}; q)_j (a; q)_m q^{nj}}{(1 - a)(aq^{n+1}; q)_j (q; q)_j} = \frac{(aq; q)_M q^{nM} (q^{1-n}; q)_M}{(q; q)_M (aq^{n+1}; q)_M}.$$

3. Show that the (α_n, β_n) defined below is a Bailey pair with $a = 1$:

$$\alpha_m = \begin{cases} -q^{6n^2-5n+1}, & m = 3n - 1 > 0, \\ q^{6n^2-n} + q^{6n^2+n}, & m = 3n > 0, \\ -q^{6n^2+5n+1}, & m = 3n + 1 > 0, \\ 1, & m = 0, \end{cases}$$

$$\beta_n = 1/(q; q)_{2n}.$$

(Slater)

4. Suppose

$$\alpha_m = \begin{cases} -q^{6n^2-2n}, & m = 3n - 1 > 0, \\ q^{6n^2-2n} + q^{6n^2+2n}, & m = 3n > 0, \\ -q^{6n^2+2n}, & m = 3n + 1 > 0, \\ 1, & m = 0, \end{cases}$$

and

$$\beta_n = q^n / (q; q)_{2n}.$$

Prove that (α_n, β_n) is a Bailey pair with $a = 1$.

(Slater)

5. Prove that

$$\alpha_n = \begin{cases} (-1)^m (q^{m(5m+1)/2} + q^{m(5m-1)/2}), & n = 2n, \\ 1, & n = 0, \\ 0, & n = \text{odd}, \end{cases}$$

$$\beta_n = \frac{1}{(q; q)_{2n}} \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix}_q q^{j^2}$$

is a Bailey pair with $a = 1$.

(Bressoud)

6. Apply (12.3.8), Exercises 10.37 and 10.39, and Jacobi's triple product formula to obtain the following six identities of Rogers:

$$(a) \quad 1 + \sum_{n=1}^{\infty} \frac{q^{n^2}}{(q; q)_{2n}} = 1 / [(q; q^2)_{\infty} (q^4; q^{20})_{\infty} (q^{16}; q^{20})_{\infty}].$$

$$(b) \quad 1 + \sum_{n=1}^{\infty} \frac{q^{n^2+2n}}{(q; q)_{2n+1}} = 1 / [(q; q^2)_{\infty} (q^8; q^{20})_{\infty} (q^{12}; q^{20})_{\infty}].$$

$$(c) \quad 1 + \sum_{n=1}^{\infty} \frac{q^{n^2+n}}{(q; q)_{2n}} = \prod_{\substack{n=1 \\ n \not\equiv \pm 1, \pm 8, \pm 9, 10 \pmod{20}}}^{\infty} (1 - q^n)^{-1}.$$

$$(d) \quad 1 + \sum_{n=1}^{\infty} \frac{q^{n^2+n}}{(q; q)_{2n+1}} = \prod_{\substack{n=1 \\ n \not\equiv \pm 3, \pm 4, \pm 7, 10 \pmod{20}}}^{\infty} (1 - q^n)^{-1}.$$

$$(e) \quad (-q^2; q^2)_{\infty} \sum_{n=0}^{\infty} \frac{q^{n^2}}{(q^4; q^4)_n} = \prod_{n=0}^{\infty} \frac{1}{(1 - q^{5n+1})(1 - q^{5n+4})}.$$

$$(f) \quad (-q^2; q^2)_{\infty} \sum_{n=0}^{\infty} \frac{q^{n^2+2n}}{(q^4; q^4)_n} = \prod_{n=0}^{\infty} \frac{1}{(1 - q^{5n+2})(1 - q^{5n+3})}.$$

Whipple's ${}_7F_6$ transformation was obtained in Chapter 7 as a byproduct of the solution of the connection coefficient problem for Jacobi polynomials. There are several q -analogues of the Jacobi polynomials. The next set of four

problems shows how the little q -Jacobi polynomials can be used to derive Watson's ${}_8\phi_7$ transformation. See Andrews and Askey [1977].

7. Define the little q -Jacobi polynomials by

$$p_n(x; \alpha, \beta : q) = {}_2\phi_1 \left(\begin{matrix} q^{-n}, \alpha\beta q^{n+1}; \\ \alpha q \end{matrix}; q, qx \right).$$

(a) Show that

$$\lim_{q \rightarrow 1^-} p_n(x; q^\alpha, q^\beta : q) = P_n^{(\alpha, \beta)}(1 - 2x) / P_n^{\alpha, \beta}(1).$$

(b) Derive the orthogonality relation

$$\begin{aligned} & \sum_{i=0}^{\infty} \frac{\alpha^i q^i (q^{i+1}; q)_{\infty}}{(\beta q^{i+1}; q)_{\infty}} p_n(q^i; \alpha, \beta : q) p_m(q^i; \alpha, \beta : q) \\ &= \begin{cases} 0 & \text{if } m \neq n, \\ \frac{\alpha^n q^n (q; q)_{\infty} (\alpha\beta q^{n+1}; q)_{\infty} (q; q)_n}{(\beta q^{n+1}; q)_{\infty} (\alpha q; q)_{\infty} (\alpha q; q)_n (1 - \alpha\beta q^{2n+1})} & \text{if } m = n. \end{cases} \end{aligned}$$

[Hint: First obtain the following identity:

$$\begin{aligned} & \sum_{i=0}^{\infty} \frac{\alpha^i q^i (q^{i+1}; q)_{\infty}}{(\beta q^{i+1}; q)_{\infty}} p_n(q^i; \alpha, \beta : q) q^{im} \\ &= \begin{cases} 0 & \text{if } 0 \leq m < n, \\ \frac{(-\alpha)^n q^{n(n+1)/2} (q; q)_{\infty} (\alpha\beta q^{2n+2}; q)_{\infty} (q; q)_n}{(\beta q^{n+1}; q)_{\infty} (\alpha q; q)_{\infty}} & \text{if } m = n. \end{cases} \end{aligned}$$

8. Suppose that

$$p_n(x; \gamma, \delta : q) = \sum_{k=0}^n a_{kn} p_k(x; \alpha, \beta : q).$$

Show that

$$\begin{aligned} a_{kn} &= \frac{(-1)^k q^{k(k+1)/2} (\gamma\delta q^{n+1}; q)_k (q^{-n}; q)_k (\alpha q; q)_k}{(\alpha\beta q^{k+1}; q)_k (q; q)_k (\gamma q; q)_k} \\ &\quad \cdot {}_3\phi_2 \left(\begin{matrix} q^{-n+k}, \gamma\delta q^{n+k+1}, \alpha q^{k+1} \\ \gamma q^{k+1}, \alpha\beta q^{2k+2} \end{matrix}; q, q \right). \end{aligned}$$

9. Note that the identity in Exercise 8 is a polynomial identity in x . Deduce that with the same a_{kn} , we have

$$\begin{aligned} & {}_{r+2}\phi_{r+1} \left(\begin{matrix} q^{-n}, \gamma\delta q^{n+1}, a_1, \dots, a_r \\ \gamma q, b_1, \dots, b_r \end{matrix}; q, qx \right) \\ &= \sum_{k=0}^n a_{kn} {}_{r+2}\phi_{r+1} \left(\begin{matrix} q^{-k}, \alpha\beta q^{k+1}, a_1, \dots, a_r \\ \alpha q, b_1, \dots, b_r \end{matrix}; q, qx \right). \end{aligned}$$

10. Take $r = 2$, $\beta = \delta$, $a_1 = \alpha q$, $x = 1$, and $b_2 = q^2 \alpha \delta a_2 / b_1$ in the previous exercise. Observe that the ${}_3\phi_2$ in a_{kn} becomes balanced and may be computed

by (10.10.3). Deduce that

$$\begin{aligned}
 {}_4\phi_3 \left(\begin{matrix} q^{-n}, \gamma \delta q^{n+1}, \alpha q, a_2 \\ \gamma q, b_1, q^2 \alpha \delta a_2 / b_1 \end{matrix}; q, q \right) &= \frac{\alpha^n q^n (\delta q; q)_n (\gamma / \alpha; q)_n}{(\alpha \delta q^2; q)_n (\gamma q; q)_n} \\
 &\cdot {}_8\phi_7 \left(\begin{matrix} q^{-n}, q \sqrt{\alpha \delta q}, -q \sqrt{\alpha \delta q}, \alpha \delta q, \alpha q, \alpha \delta q^2 / b_1, \gamma \delta q^{n+1}, b_1 / a_2 \\ \sqrt{\alpha \delta q}, -\sqrt{\alpha \delta q}, \delta q, b_1, \alpha q^{-n+1} / \gamma, \alpha \delta a_2 q^2 / b_1, \alpha \delta q^{n+2} \end{matrix}; q, a_2 / \gamma \right).
 \end{aligned}$$

The standard form of Watson's transformation is obtained from this formula after proper identification.

11. The big q -Jacobi polynomials are defined by

$$\begin{aligned}
 P_n^{(\alpha, \beta)}(x; c, d : q) &= c^n \frac{q^{-(\alpha+1)n} (q^{\alpha+1}; q)_n (-q^{\alpha+1} d/c; q)_n}{(q; q)_n (-q; q)_n} \\
 &\cdot {}_3\phi_2 \left(\begin{matrix} q^{-n}, q^{n+\alpha+\beta+1}, x q^{\alpha+1} d/c \\ q^{\alpha+1}, -q^{\alpha+1} d/c \end{matrix}; q, q \right).
 \end{aligned}$$

Prove that the orthogonality relation is

$$\begin{aligned}
 \int_{-d}^c P_n^{(\alpha, \beta)}(x; c, d : q) P_m^{(\alpha, \beta)}(x; c, d : q) \frac{(qx/c; q)_\infty (-qx/d; q)_\infty d_q x}{(q^{\alpha+1} x/c; q)_\infty (-q^{\beta+1} x/d; q)_\infty} \\
 = \begin{cases} 0 & m \neq n, \\ \frac{(cd)^n q^{n(n-1)/2} (q^{\alpha+1}; q)_n (q^{\beta+1}; q)_n}{(q^{\alpha+\beta+1}; q)_n (q; q)_n (1 - q^{2n+\alpha+\beta+1})} \\ \cdot \frac{(-q^{\beta+1} c/d; q)_n (-q^{\alpha+1} d/c; q)_n (1 - q^{\alpha+\beta+1}) M}{(-q; q)_n (-q; q)_n}, & m = n, \end{cases}
 \end{aligned}$$

where

$$M = \frac{cd(1-q)(q; q)_\infty (q^{\alpha+\beta+2}; q)_\infty (-d/c; q)_\infty (-c/d; q)_\infty}{(c+d)(q^{\alpha+1}; q)_\infty (q^{\beta+1}; q)_\infty (-q^{\alpha+1} d/c; q)_\infty (-q^{\beta+1} c/d; q)_\infty}.$$

Note that when $c = d = 1$, the weight function in the q -integral tends to $(1-x)^\alpha (1+x)^\beta$ as $q \rightarrow 1^-$.

12. A set of q -Laguerre polynomials is defined by

$$L_n^{(\alpha)}(x; q) = \frac{(q^{\alpha+1}; q)_n}{(q; q)_n} \sum_{k=0}^n \frac{(q^{-n}; q)_k q^{\binom{k}{2}} (1-q)^k (q^{n+\alpha+1} x)^k}{(q^{\alpha+1}; q)_k (q; q)_k}.$$

(a) Replace x with $-(1-q)q^{-(\beta+1)}x$ and let $\beta \rightarrow -\infty$ in the little q -Jacobi polynomial of degree n . Show that the result is

$$\frac{(q; q)_n}{(q^{\alpha+1}; q)_n} L_n^{(\alpha)}(x; q).$$

(b) Prove that $\lim_{q \rightarrow 1^-} L_n^{(\alpha)}(x; q) = L_n^\alpha(x)$.

(c) Prove the discrete orthogonality relation

$$\sum_{k=-\infty}^{\infty} L_n^{(\alpha)}(q^k; q) L_m^{(\alpha)}(q^k; q) \frac{q^{k\alpha+k}}{(-(1-q)q^k; q)_{\infty}} \\ = \begin{cases} 0, & m \neq n, \\ \frac{(q^{\alpha+1}; q)_n}{q^n(q; q)_n} \sum_{k=-\infty}^{\infty} \frac{q^{k\alpha+k}}{(-(1-q)q^k; q)_{\infty}} & m = n. \end{cases}$$

(Use Ramanujan's ${}_1\psi_1$ sum.)

(d) Prove the continuous orthogonality relation

$$\int_0^{\infty} L_m^{(\alpha)}(x; q) L_n^{(\alpha)}(x; q) \frac{x^{\alpha}}{(-(1-q)x; q)_{\infty}} dx \\ = \begin{cases} 0, & m \neq n, \\ \frac{\Gamma(\alpha+1)\Gamma(-\alpha)(q^{n+1}, q)_n}{\Gamma_q(-\alpha)(q; q)_n q^n}, & m = n. \end{cases}$$

(Moak)

13. A multiplicative shift of the variable x in Exercise 12 gives a set of q -Laguerre polynomials defined by

$$L_n^{\alpha}(x; q) = \sum_{k=0}^n \frac{(q^{-n}; q)_k q^{nk+(\alpha-\beta)k+k(k+1)/2} x^k}{(q^{\alpha+1}; q)_k (q; q)_k}.$$

(a) Show that

$$\int_0^{\infty} \frac{L_n^{\alpha}(x; q) L_m^{\alpha}(x; q) x^{\alpha-\beta} (-q^{\beta+1} x^{-1}; q)_{\infty}}{(-x; q)_{\infty} (-qx^{-1}; q)_{\infty}} dx \\ = \begin{cases} \frac{q^{-n}(q; q)_n (q; q)_{\infty}}{(q^{\alpha+1}; q)_n (q^{\alpha+1}; q)_n} \frac{\Gamma(\alpha+1-\beta)\Gamma(\beta-\alpha)}{\Gamma_q(\alpha+1-\beta)\Gamma_q(\beta-\alpha)}, & m = n, \\ 0, & m \neq n. \end{cases}$$

(b) Set $\alpha - \beta = c$ (fixed c) and let $\alpha \rightarrow \infty$ in $L_n^{\alpha}(x; q)$. Show that the result is the Stieltjes–Wigert polynomial

$$S_n(x; q) = \sum_{k=0}^n \frac{q^{k^2} (-q^c x)^k}{(q; q)_k (q; q)_{n-k}}.$$

(c) Prove that a weight function for the Stieltjes–Wigert polynomials is

$$\omega(x) = \frac{x^c}{(-x; q)_{\infty} (-q/x; q)_{\infty}}.$$

For Exercises 12 and 13, see Gasper and Rahman [1990, Chapter 7].

A

Infinite Products

A.1 Infinite Products

For readers unfamiliar with infinite products, a brief introduction is given here.

Definition A.1.1 Let $p_{n,k} = \prod_{m=k}^n (1+a_m)$. If there is a k for which $p_{n,k}$ converges to a nonzero value p as $n \rightarrow \infty$, then we say that the infinite product $\prod_{n=1}^{\infty} (1+a_n)$ converges. We write it as $p = \prod_{n=k}^{\infty} (1+a_n)$. The reason for not taking $k = 1$ is to allow an finite number of zero factors. The convergence of the product is said to be absolute if $\prod_{n=1}^{\infty} (1+|a_n|)$ converges.

The following basic theorem reduces the problem of convergence of a product to that of a series. For simplicity, assume that $\operatorname{Re} a_n > -1$, $n = 1, 2, \dots$. If not, start the product after this holds.

Theorem A.1.2 The product $\prod_{n=1}^{\infty} (1+a_n)$ converges if and only if the series $\sum_{n=1}^{\infty} \log(1+a_n)$ converges.

Proof. Suppose that $S_n = \sum_{m=1}^n \log(1+a_m)$ converges to S . Since $\exp$ is a continuous function,

$$\prod_{m=1}^n (1+a_m) = \exp(S_n) \quad \text{converges to } e^S = \prod_{n=1}^{\infty} (1+a_n).$$

To prove the converse, let $1+a_m = A_m e^{i\theta_m}$ and $\prod_{m=1}^n (1+a_m) = B_n e^{i\phi_n}$. The convergence of $\prod (1+a_n)$ implies $a_m \rightarrow 0$ so that $\theta_m \rightarrow 0$ and ϕ_n can be chosen so that, say, $\phi_n \rightarrow \phi$. We also have

$$\theta_1 + \theta_2 + \dots + \theta_n = \phi_n + 2\pi k_n,$$

where k_n is an integer. Therefore,

$$\theta_{n+1} = \phi_{n+1} - \phi_n + 2\pi(k_{n+1} - k_n).$$

Since $k_{n+1} - k_n$ is an integer, $\theta_{n+1} \rightarrow 0$, and since $\phi_{n+1} - \phi_n \rightarrow 0$, we must have $k_n = k$, a constant for sufficiently large n . Thus, for sufficiently large n ,

$$S_n = \log p_n + 2\pi ki.$$

Let $n \rightarrow \infty$ to get

$$S = \log p + 2\pi ki.$$

This proves the theorem. For absolute convergence the condition is simpler and is contained in the next theorem. ■

Theorem A.1.3 *The product $\prod_{n=1}^{\infty} (1 + a_n)$ converges absolutely if and only if $\sum_{n=1}^{\infty} a_n$ converges absolutely.*

Proof. Convergence of the series or the product implies that $a_n \rightarrow 0$. For sufficiently large n we must have $|a_n| \leq 1/2$. Suppose $a_n \neq 0$ and n large. Then

$$\begin{aligned} \left| 1 - \frac{\log(1 + a_n)}{a_n} \right| &= \left| \frac{a_n}{2} - \frac{a_n^2}{3} + \cdots \right| \\ &\leq \frac{1}{2} [|a_n| + |a_n|^2 + \cdots] \\ &= \frac{1}{2} \frac{|a_n|}{1 - |a_n|} \leq \frac{1}{2}. \end{aligned}$$

So

$$-\frac{1}{2} \leq \left| \frac{\log(1 + a_n)}{a_n} \right| - 1 \leq \frac{1}{2}$$

or

$$\frac{1}{2}|a_n| \leq |\log(1 + a_n)| \leq \frac{3}{2}|a_n|.$$

These inequalities together with Theorem A.1.2 imply the result and also the fact that absolute convergence of a product implies its convergence. ■

Definition A.1.4 *The infinite product*

$$\prod_{n=1}^{\infty} (1 + a_n(x)),$$

where x is a real or complex variable in a domain, is uniformly convergent if

$$p_n(x) = \prod_{m=k}^n (1 + a_m(x))$$

converges uniformly in that domain, for each k .

Theorem A.1.5 *If the series $\sum_{n=1}^{\infty} |a_n(x)|$ converges uniformly in some region, then the product $\prod_{n=1}^{\infty} (1 + a_n(x))$ also converges uniformly in that region.*

The proof of this result is left to the reader.

Corollary A.1.6 *If $a_n(x)$ is analytic in some region of the complex plane and $\prod_{n=1}^{\infty} (1 + a_n(x))$ converges uniformly in that region, then the infinite product represents an analytic function in that region.*

Exercises

1. Prove directly from the definition that the product $(1 - \frac{1}{2})(1 + \frac{1}{3})(1 - \frac{1}{4}) \cdots$ is convergent.
2. Prove that an absolutely convergence product is convergent.
3. Show that if $\sum a_n, \sum a_n^2, \dots, \sum a_n^{k-1}, \sum |a_n|^k$ are all convergent, then $\prod (1 + a_n)$ is convergent.
4. Discuss the convergence of the product

$$\prod_{n=1}^{\infty} \left\{ \left(1 + \frac{1}{n}\right)^x \left(1 - \frac{x}{n}\right) \right\}.$$

B

Summability and Fractional Integration

B.1 Abel and Cesàro Means

The following theorem, the first part of which was proved by Abel, is often encountered in a first course in real analysis.

Theorem B.1.1 *Suppose that the series $\sum_{n=0}^{\infty} b_n$ converges to B . Then $\sum_{n=0}^{\infty} b_n x^n$ converges uniformly on $[0, 1]$, and in particular*

$$\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} b_n x^n = B. \quad (\text{B.1.1})$$

Moreover, if $B_n = \sum_{k=0}^n b_k$, then

$$\lim_{n \rightarrow \infty} \frac{B_0 + B_1 + \cdots + B_n}{n+1} = B. \quad (\text{B.1.2})$$

The example of the series $1 - 1 + 1 - \cdots$ shows that the limits in (B.1.1) and (B.1.2) may exist even when the series does not converge. Thus (B.1.1) and (B.1.2) may be used to assign a sum to divergent series.

The series $\sum_{n=0}^{\infty} b_n$ is Abel summable or summable A to B if (B.1.1) holds; and it is Cesàro summable or summable $(C, 1)$ to B if (B.1.2) holds. We also speak of (B.1.1) as the Abel mean and (B.1.2) as the Cesàro mean of the series $\sum_{n=0}^{\infty} b_n$.

The Abel and Cesàro means of the series $1 - 1 + 1 - \cdots$ are both $1/2$. The next theorem shows that Cesàro summability is a stronger requirement than Abel summability.

Theorem B.1.2 *If the series $\sum_{n=0}^{\infty} b_n$ is $(C, 1)$ summable to B , then it is A summable to B .*

Proof. Use summation by parts to get

$$\begin{aligned}
 \sum_{n=0}^{\infty} b_n x^n - B &= (1-x) \sum_{n=0}^{\infty} B_n x^n - B \left(\text{where } B_n = \sum_{k=0}^n b_k \right) \\
 &= (1-x)^2 \sum_{n=0}^{\infty} (B_0 + \cdots + B_n) x^n - B \\
 &= (1-x)^2 \sum_{n=0}^{\infty} \{(B_0 + \cdots + B_n) - (n+1)B\} x^n. \quad (\text{B.1.3})
 \end{aligned}$$

Now (B.1.2) implies that, for $\epsilon > 0$, there exists an integer N such that

$$|(B_0 + \cdots + B_n) - (n+1)B| < \epsilon(n+1) \quad \text{for } n \geq N.$$

Use this in (B.1.3) to arrive at the necessary result. ■

The example of the series $1 - 2 + 3 - \cdots$ shows that the Abel mean may exist (in this case it is $1/4$) but the Cesàro mean may not. The Abel and Cesàro means, together with other summability methods, are very useful in analysis and analytic number theory. As an elementary example, consider the following theorem of Abel on the product of two series:

Theorem B.1.3 Suppose $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ are convergent series. Let $c_n = \sum_{k=0}^n a_k b_{n-k}$ and suppose that $\sum_{n=0}^{\infty} c_n$ is convergent. Then

$$\sum_{n=0}^{\infty} c_n = \sum_{n=0}^{\infty} a_n \sum_{n=0}^{\infty} b_n. \quad (\text{B.1.4})$$

Proof. Since

$$\begin{aligned}
 c_n &= \sum_{k=0}^n a_{n-k} b_k, \\
 \sum_{k=0}^{\infty} c_n r^n &= \sum_{j=0}^{\infty} a_j r^j \sum_{k=0}^{\infty} b_k r^k
 \end{aligned}$$

and Theorem B.1.1 completes the proof. ■

The Cesàro and Abel means play a very significant role in the theory of Fourier series and transforms. We take a very brief look at how they appear in Fourier series since these ideas will play a role elsewhere in the book.

Suppose $f(x)$ is an integrable function of period 2π . Let $\sum_{-\infty}^{\infty} a_n e^{inx}$ be its Fourier series. An important question is: When does the Fourier series of a function converge to the function? Once again it is easier to deal with the Cesàro mean or

the Abel mean. The Abel sum is given by

$$\sum_{-\infty}^{\infty} a_n r^{|n|} e^{inx}, \quad |r| < 1. \quad (\text{B.1.5})$$

Since

$$a_n = \frac{1}{2\pi} \int_0^{2\pi} f(t) e^{-int} dt,$$

the sum in (B.1.5) is equal to

$$\frac{1}{2\pi} \int_0^{2\pi} f(t) \sum_{-\infty}^{\infty} r^{|n|} e^{in(x-t)} dt.$$

The sum inside the integral is called the Poisson kernel and it is a straightforward calculation to show that it is equal to

$$\frac{1 - r^2}{1 - 2r \cos(x - t) + r^2} \equiv P_r(x - t). \quad (\text{B.1.6})$$

The following properties of $P_r(x)$ are worth noting:

- (i) $P_r(x) \geq 0$,
- (ii) $\frac{1}{2\pi} \int_0^{2\pi} P_r(x) dx = 1$, and
- (iii) for $\delta > 0$, $\max_{\delta \leq x \leq 2\pi - \delta} P_r(x) \rightarrow 0$ as $r \rightarrow 1^-$.

These properties can be used to give a proof of the following theorem:

Theorem B.1.4 *If f is periodic and integrable on $(0, 2\pi)$ then the Abel means of the Fourier series converge to $\frac{1}{2}\{f(x_0+) + f(x_0-)\}$ at every point x_0 where the right and left limits $f(x_0\pm)$ exist.*

A similar result exists for Cesàro means. The n th partial sum of the Fourier series is given by

$$s_n(x) = \sum_{k=-n}^n a_k e^{ikx} = \frac{1}{2\pi} \int_0^{2\pi} \sum_{k=-n}^n e^{ik(x-t)} f(t) dt.$$

The sum inside the integral is

$$1 + 2 \sum_{k=0}^n \cos k(x - t) = \frac{\sin(n + 1/2)(x - t)}{\sin((x - t)/2)}. \quad (\text{B.1.7})$$

This expression is called the Dirichlet kernel. One drawback of the Dirichlet kernel is that it is not always positive. However, if we take the $(C, 1)$ mean of the Fourier

series, then we get

$$\begin{aligned}\sigma_n(x) &= \frac{s_0(x) + s_1(x) + \cdots + s_n(x)}{n+1} \\ &= \frac{1}{2\pi(n+1)} \int_0^{2\pi} f(t) \sum_{k=0}^n \frac{\sin(n + \frac{1}{2})(x-t)}{\sin((x-t)/2)} dt.\end{aligned}$$

The sum in the integral denoted by $K_n(x-t)$ is called the Fejér kernel. The sum is equal to

$$\left\{ \frac{\sin \frac{1}{2}(n+1)(x-t)}{\sin \frac{1}{2}(x-t)} \right\}^2 \geq 0. \quad (\text{B.1.8})$$

The Fejér kernel has the three properties of the Poisson kernel mentioned above. One may then deduce the following theorem:

Theorem B.1.5 *For f as in Theorem B.1.4, the Cesàro means of the Fourier series of f converge to $\frac{1}{2}\{f(x_0+) + f(x_0-)\}$ at every point x_0 where the right and left limits $f(x_0\pm)$ exist.*

If we assume f to be continuous on $[0, 2\pi]$, then the Abel and Cesàro means converge to f . An important consequence of Theorem B.1.5 is the following corollary:

Corollary B.1.6 *A continuous function on $[0, 2\pi]$ can be uniformly approximated by trigonometric polynomials, that is, polynomials of the form $\sum_{-n}^n a_k e^{ikx}$.*

Since e^{ikx} is uniformly approximated on $[0, 2\pi]$ by partial sums of its Taylor series, we get another proof (see Exercise 1.40) of the Weierstrass approximation theorem.

B.2 Cesàro Means (C, α)

In the previous section, we saw that the $(C, 1)$ means do not assign a value to the sum $1 - 2 + 3 - 4 + \cdots$. To handle this and some other situations, we define Cesàro means of higher order.

For the series $\sum_0^\infty b_n$, set

$$B_n^{(0)} = B_0 + B_1 + \cdots + B_n,$$

where

$$B_n = \sum_{k=0}^n b_k.$$

The limit of the sequence $\{\frac{B_n^{(0)}}{n+1}\}$ gives the $(C, 1)$ mean. Now, set

$$B_n^{(1)} = B_0^{(0)} + B_1^{(0)} + \cdots + B_n^{(0)} \quad (\text{B.2.1})$$

and

$$E_n^{(1)} = 1 + 2 + \cdots + n + 1 = \frac{(n+1)(n+2)}{2}.$$

The limit of the sequence $\{\frac{B_n^{(1)}}{E_n^{(1)}}\}$ is the $(C, 2)$ mean of the series $\sum_0^\infty b_n$.

Similarly,

$$B_n^{(2)} = B_0^{(1)} + B_1^{(1)} + \cdots + B_n^{(1)}$$

and

$$E_n^{(2)} = 1 + 3 + \cdots + \frac{(n+1)(n+2)}{2} = \frac{(n+1)(n+2)(n+3)}{3!}.$$

Thus, the $(C, 3)$ mean is given by

$$\lim_{n \rightarrow \infty} (B_n^{(2)} / E_n^{(2)}).$$

Define $B_n^{(k)}$ and $E_n^{(k)}$ inductively by

$$B_n^{(k)} = B_0^{(k-1)} + B_1^{(k-1)} + \cdots + B_n^{(k-1)}$$

and

$$E_n^{(k)} = E_0^{(k-1)} + E_1^{(k-1)} + \cdots + E_n^{(k-1)}.$$

It is possible to express $B_n^{(k)}$ explicitly in terms of b_n as follows. Note that

$$\sum_{n=0}^{\infty} B_n^{(k)} x^n = (1-x)^{-1} \sum_{n=0}^{\infty} B_n^{(k-1)} x^n = \cdots = (1-x)^{-(k+1)} \sum_{n=0}^{\infty} b_n x^n$$

and

$$\sum_{n=0}^{\infty} E_n^{(k)} x^n = (1-x)^{-(k+1)}.$$

So

$$B_n^{(k)} = \sum_{\ell=0}^n \left(\frac{(k+1)_\ell}{\ell!} \right) b_{n-\ell} \quad (\text{B.2.2})$$

and

$$E_n^{(k)} = \frac{(k+1)_n}{n!}. \quad (\text{B.2.3})$$

We define the (C, k) mean of $\sum_0^\infty b_n$ as the limit of the $B_n^{(k)} / E_n^{(k)}$ as $n \rightarrow \infty$. Note that these quotients are meaningful even when k is not a positive integer. We take k to be real and greater than -1 .

Definition B.2.1 The series $\sum_0^\infty b_n$ is (C, α) summable to B (for $\alpha > -1$) if

$$\lim_{n \rightarrow \infty} \frac{n!}{(\alpha + 1)_n} \sum_{\ell=0}^n \frac{(\alpha + 1)_k}{\ell!} b_{n-\ell} = B. \quad (\text{B.2.4})$$

Remark B.2.1 Note that the limit in (B.2.4) is equal to

$$\lim_{n \rightarrow \infty} \frac{\Gamma(\alpha + 1)}{n^\alpha} B_n^{(\alpha)}.$$

B.3 Fractional Integrals

It is possible to extend the summability definitions to integrals. Suppose $f(t)$ is integrable on finite intervals $(0, x)$. We say, for example, that f is Abel integrable, that is, $\int_0^\infty f(t)dt$ exists in the Abel sense, if

$$\lim_{\lambda \rightarrow 0^+} \int_0^\infty e^{-\lambda t} f(t)dt \text{ exists.}$$

We also write this as

$$\int_0^\infty f(t)dt = \lim_{\lambda \rightarrow 0^+} \int_0^\infty e^{-\lambda t} f(t)dt \quad (\text{Abel}).$$

To define Cesàro integrability (C, k) , where k is an integer, we first have the integral analog of $B_n^{(k)}$ (see (B.2.2)) as

$$f_{(k)}(x) = \int_0^x f_{(k-1)}(t)dt = \int_0^x (x-t)f_{(k-2)}(t)dt = \cdots = \frac{1}{k!} \int_0^x (x-t)^k f(t)dt. \quad (\text{B.3.1})$$

Following the remark after (B.2.4) we say that $\int_0^\infty f(t)dt$ is (C, k) integrable if

$$\lim_{x \rightarrow \infty} \frac{\Gamma(k+1)}{x^k} f_k(x) = \lim_{x \rightarrow \infty} \int_0^x \left(1 - \frac{t}{x}\right)^k f(t)dt \text{ exists.}$$

Observe that the final expression for $f_k(x)$ in (B.3.1) is meaningful for all real $k > -1$. Thus we have a definition of (C, α) integrability for $\alpha > -1$.

Note also that the formula for $f_{(k)}$ in (B.3.1) expresses $f_{(k)}$ as a $(k+1)$ -fold integral of f . Thus, we define $I_\alpha f$, the α -fold integral of f for $\text{Re } \alpha > 0$, by the formula

$$I_\alpha f(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t)dt. \quad (\text{B.3.2})$$

It is easy to check that the operator I_α satisfies the relation

$$I_\alpha I_\beta = I_{\alpha+\beta}, \quad \operatorname{Re}(\alpha, \beta) > 0.$$

The operator I_α is called a fractional integral of order α . Euler's integral expression for the ${}_2F_1$ hypergeometric series can now be interpreted as a fractional integral. Write the formula as

$$\begin{aligned} {}_2F_1\left(\begin{matrix} a, b \\ c \end{matrix}; x\right) &= \frac{\Gamma(c)x^{1-c}}{\Gamma(b)\Gamma(c-b)} \int_0^x t^{b-1}(x-t)^{c-b-1}(1-t)^{-a} dt \\ &= \frac{\Gamma(c)}{\Gamma(b)} x^{1-c} (I_{c-b} f)(x), \end{aligned}$$

where $f(t) = t^{b-1}(1-t)^{-a}$.

Remark B.3.1 Abel and Cesàro integrability can be used to study Fourier integrals just as the corresponding summability is used to study Fourier series.

B.4 Historical Remarks

Euler had used Abel means and other methods for associating a sum to a divergent series. In particular, he obtained the functional equation of the zeta function in the form

$$\frac{1 - 2^{s-1} + 3^{s-1} - \dots}{1 - 2^{-s} + 3^{-s} - \dots} = -\frac{(s-1)!(2^s-1)}{(2^{s-1}-1)\pi^s} \cos \frac{1}{2}s\pi.$$

When $\operatorname{Re} s > 1$, the series in the denominator converges but the series in the numerator is interpreted in the Abelian sense. It can be proved that

$$\lim_{x \rightarrow 1^-} \sum_{n=1}^{\infty} (-1)^{n-1} n^{s-1} x^n = \zeta(1-s)(2^s-1);$$

thus Euler had the functional equation correct. He verified it only for integer values of s and for $s = \frac{1}{2}$ and $\frac{3}{2}$. Abel's name is attached to this method of summation since he proved formula (B.1.1).

Leibniz used $(C, 1)$ means to evaluate $1 - 1 + 1 - \dots$ and somewhat later D. Bernoulli employed the same method to consider the more general series $\sum_{n=0}^{\infty} b_n$, where $b_{n+p} = b_n$ for all n and $\sum_{k=0}^{p-1} b_k = 0$. Neither Leibniz nor Bernoulli explicitly stated that he was giving a new definition of convergence. This was done by Cesàro. For more on the history of Abel and Cesàro means the reader should consult the very interesting treatment given in Hardy [1949].

Abel was apparently the first mathematician to use fractional calculus, though the concept had been considered by others before him. Abel used fractional calculus in his solution to the problem of finding the tautochrone – the curve down which

a particle slides freely under gravity with zero initial velocity and reaches the bottom in the same amount of time regardless of the starting point, provided that the starting and lowest points are distinct. This problem had already been solved by Huygens; the motivation was the construction of a clock in which the period of oscillation does not depend on the amplitude of the pendulum.

Suppose that the particle slides down from the point (a, b) and that the bottom of the curve is the origin of coordinates. If s is the length measured from the origin, then conservation of energy implies that

$$\text{velocity} = v = ds/dt = \sqrt{2g(b-y)}$$

at a point on the curve with ordinate y . Set $ds/dy = f(y)$ to see that the time taken to reach the bottom is given by

$$T(b) = \frac{1}{\sqrt{2g}} \int_0^b \frac{f(y)}{\sqrt{b-y}} dy.$$

Abel observed that

$$T(b) = \sqrt{\frac{\pi}{2g}} (I_{1/2} f)(b).$$

Since $T(b)$ is independent of b , use (B.3.2) to get

$$I_{1/2}(I_{1/2} f)(b) = \frac{\sqrt{2g}T}{\pi} \int_0^b \frac{dy}{\sqrt{b-y}} = \frac{2T}{\pi} \sqrt{2gb}.$$

Now take the derivative of both sides to obtain

$$f(b) = \frac{T}{\pi} \sqrt{\frac{2g}{b}}.$$

Since

$$f(y) = \frac{ds}{dy} = \left[1 + \left(\frac{dx}{dy} \right)^2 \right]^{1/2},$$

we get a differential equation that can be solved. The tautochrone curve turns out to be a cycloid. An English translation of Abel's [1826] paper on this topic is also available.

The above calculations suggest that the concept of a fractional derivative would also be useful. For smooth enough functions, in particular analytic functions, it is possible to define fractional derivatives as follows:

$$D_\alpha f(x) = \frac{d^n}{dx^n} I_{n-\alpha} f(x), \quad (\text{B.4.1})$$

where $0 < \text{Re } \alpha < n$, and n is an integer. Observe that we took the integral first and then the derivative. If the calculations are done in the opposite order, a different

function arises since the first $n - 1$ terms are removed by the n th derivative. For an analytic function

$$f(x) = \sum_0^{\infty} a_k x^k$$

the fractional integral has the form

$$I_{\alpha} f(x) = \sum_0^{\infty} \frac{k!}{\Gamma(k + \alpha + 1)} a_k x^{k+\alpha},$$

as is easily verified. This shows that the value of n in (B.4.1) may be arbitrarily chosen as long as $n > \operatorname{Re} \alpha$. In fact,

$$D_k I_{\alpha} = D_n I_{\alpha+n-k}.$$

However, the inclusion of fractional derivatives with fractional integrals does not give a group, for although

$$D_{\alpha} D_{\beta} = D_{\alpha+\beta}$$

when α, β and $\alpha + \beta$ are not integers,

$$D_{1/2} D \neq D_{3/2}$$

for the reason mentioned above.

Exercises

1. Prove Theorem B.1.1.
2. With the notation of Theorem B.1.3, prove that if $\sum_0^{\infty} a_n$ converges absolutely and $\sum_0^{\infty} b_n$ converges, then $\sum_0^{\infty} c_n$ converges.
3. Prove that if the series $\sum_0^{\infty} b_n$ is (C, α) summable to B , and $\beta > \alpha$, then the series is (C, β) summable to B .
4. Prove the following extension of Theorem B.1.2: If the series $\sum_0^{\infty} b_n$ is (C, α) summable to B , then it is Abel summable to B .
5. Show that

$$\sum_{n=1}^{\infty} (-1)^{n-1} n^m e^{-ny} = (-1)^m \frac{d^m}{dy^m} (1 + e^y)^{-1}.$$

Deduce that

$$1 - 2^{2k} + 3^{2k} = \dots = 0,$$

$$1 - 2^{2k-1} + 3^{2k-1} - \dots = (-1)^{k-1} \frac{2^{2k} - 1}{2k} B_k$$

in the sense of Abel summability. This result is due to Euler. See the historical introduction in Hardy [1949].

6. A sequence S_n converges to S in the sense of Borel if

$$\lim_{t \rightarrow \infty} e^{-t} \sum_{n=0}^{\infty} \frac{S_n}{n!} t^n = S.$$

Show that if S_n converges to S then $S_n \rightarrow S$ in the Borel sense. Also prove that if $S_n(x) = 1 + x + \cdots + x^n$, then $S_n(x) \rightarrow 1/(1-x)$ in the Borel sense for $\operatorname{Re} x < 1$.

7. Show that if $\beta \geq \alpha \geq -1$ and the (C, β) means of Σa_n are nonnegative, then the (C, α) means of $\Sigma a_n r^n$ are nonnegative for $0 \leq r \leq (\alpha + 1)/(\beta + 1)$.

Hint: Observe that

$$(1 - \omega)^{-\alpha-1} \Sigma a_n r^n \omega^n = (1 - \omega)^{-\alpha-1} (1 - r\omega)^{\beta+1} (1 - r\omega)^{-\beta-1} \Sigma a_n r^n \omega^n$$

and show that the power series coefficients of $(1 - \omega)^{-\alpha-1} (1 - r\omega)^{\beta+1}$ and those of $(1 - r\omega)^{-\beta-1} \Sigma a_n r^n \omega^n$ are nonnegative. This proof is due to Bustoz [1974].

8. A series $\sum_1^{\infty} a_n$ is summable (L) (Lambert summable) to S if

$$\lim_{x \rightarrow 1^-} (1 - x) \sum_1^{\infty} \frac{n a_n x^n}{1 - x^n} = S.$$

Prove that if $\Sigma a_n = S(C, \alpha)$, for some α , then $\Sigma a_n = S(L)$.

9. Prove Dirichlet's theorem that if χ is a quadratic Dirichlet character, then $L(1, \chi) = \sum_{n=1}^{\infty} \chi(n)/n \neq 0$ as follows:

(a) Show that $\sum_{d|n} \chi(d) \geq 0$.

(b) Show that

$$\lim_{x \rightarrow 1^-} \sum_{n=1}^{\infty} \frac{\chi(n) x^n}{1 - x^n} = \lim_{x \rightarrow 1^-} \sum_{n=1}^{\infty} \left(\sum_{d|n} \chi(d) \right) x^n = \infty.$$

(c) Let

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{\chi(n)}{n(1-x)} - \frac{\chi(n)x^n}{1-x^n} \right) =: \sum_{n=1}^{\infty} \frac{\chi(n)}{1-x} b_n.$$

Show that $b_1 \geq b_2 \geq \cdots \geq b_n \geq b_{n+1} \geq \cdots$.

(d) Note that $|\sum_{n=1}^m \chi(n)| \leq M$, where M is a constant that does not depend on m . Use summation by parts to prove that $f(x) \leq \frac{M b_1}{1-x} = M$.

(e) Deduce that $L(1, \chi) \neq 0$.

This proof is taken from Monsky [1994].

10. Prove the following theorem of Hardy and Littlewood: If $a_n \geq 0$ and

$$\lim_{x \rightarrow 1^-} (1 - x) \sum_{n=0}^{\infty} a_n x^n = 1,$$

then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^n a_k = 1.$$

Karamata's proof is sketched below.

(a) Show that

$$\lim_{x \rightarrow 1^-} (1-x) \sum_{n=0}^{\infty} a_n x^n g(x^n) = \int_0^1 g(t) dt,$$

when $g(t)$

- (i) is a polynomial,
 - (ii) is a continuous function, or
 - (iii) has a discontinuity of the first kind.
- (b) Take $g(t) = 0$ ($0 \leq t < 1/e$) = $1/t$ ($1/e \leq t \leq 1$) and complete the proof of the theorem.

C

Asymptotic Expansions

C.1 Asymptotic Expansion

Let x be a real or complex variable in an unbounded region D and let $\sum_0^\infty a_n x^{-n}$ be a formal power series that may be convergent or divergent.

Definition C.1.1 *The series $\sum_0^\infty a_n x^{-n}$ is an asymptotic expansion of a function $f(x)$ if*

$$f(x) = \sum_0^{n-1} a_k x^{-k} + R_n(x), \quad (\text{C.1.1})$$

where $R_n(x) = o(x^{-n})$ as $x \rightarrow \infty$ in D . Usually (C.1.1) is written as

$$f(x) \sim a_0 + a_1 x^{-1} + a_2 x^{-2} + \cdots \quad \text{as } x \rightarrow \infty \text{ in } D. \quad (\text{C.1.2})$$

This definition is due to Poincaré [1886]. As a simple example we have the following asymptotic expansion of $(1+x)^{-1}$:

$$\frac{1}{1+x} \sim \frac{1}{x} - \frac{1}{x^2} + \frac{1}{x^3} - \cdots \quad \text{as } x \rightarrow \infty. \quad (\text{C.1.3})$$

In this case the series is convergent. The more interesting situation occurs when the series is divergent. Consider the complementary error function of a real variable $x \geq 0$,

$$\operatorname{erfc} x = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt. \quad (\text{C.1.4})$$

Successive integration by parts gives

$$\begin{aligned} \operatorname{erfc} x &= \frac{2}{\sqrt{\pi}} \left[\frac{e^{-x^2}}{2x} - \int_x^\infty \frac{e^{-t^2}}{2t^2} dt \right] \\ &= \frac{e^{-x^2}}{\sqrt{\pi} x} \left[1 + \sum_{k=1}^n (-1)^k \frac{1 \cdots 3 \cdot (2k-1)}{(2x^2)^k} + R_n(x) \right], \end{aligned}$$

where

$$R_n(x) = (-1)^{n+1} \frac{1 \cdot 3 \cdots (2n+1)}{2^n} x e^{x^2} \int_x^\infty \frac{e^{-t^2}}{t^{2n+2}} dt.$$

It is easy to see that

$$|R_n(x)| \leq \frac{1 \cdot 3 \cdots (2n+1)}{(2x^2)^{n+1}}. \quad (\text{C.1.5})$$

Hence

$$\frac{e^{-x^2}}{\sqrt{\pi}x} \left[1 + \sum_{k=1}^{\infty} (-1)^k \frac{1 \cdot 3 \cdots (2k-1)}{(2x^2)^k} \right]$$

is an asymptotic expansion of the complementary error function. It is clear that the series diverges for all $x > 0$. However, if a fixed number of terms is taken, then for large enough x a good approximation of $\text{erfc } x$ is obtained. On the other hand, for a given x , taking more and more terms of the series does not improve the approximation, since the series diverges.

C.2 Properties of Asymptotic Expansions

The following theorem follows almost immediately from the definition.

Theorem C.2.1 *A function $f(x)$ has an asymptotic expansion $\Sigma a_n x^{-n}$ if and only if for each n ,*

$$x^n \left[f(x) - \sum_{k=0}^{n-1} a_k x^{-k} \right] \rightarrow a_n \quad \text{as } x \rightarrow \infty \text{ in } D, \quad (\text{C.2.1})$$

uniformly with respect to arg x when x is complex.

Proof. It is easy to see that (C.2.1) implies (C.1.2). To reason in the other direction, observe that

$$x^n R_n(x) = x^n [a_n x^{-n} + R_{n+1}(x)] \rightarrow a_n \quad \text{as } x \rightarrow \infty,$$

and the theorem is proved. ■

A consequence of this theorem is that a function has at most one asymptotic expansion in D . In a different unbounded region the asymptotic expansion may be different. However, two different functions may have the same asymptotic expansion in some region. For example, the function e^{-x} in $|\arg x| \leq \frac{1}{2}\pi - \delta < \frac{1}{2}\pi$ satisfies $x^n e^{-x} \rightarrow 0$ as $x \rightarrow \infty$. By (C.2.1),

$$e^{-x} \sim 0 + \frac{0}{x} + \frac{0}{x^2} + \cdots \quad |\arg x| \leq \frac{1}{2}\pi - \delta, \quad x \rightarrow \infty.$$

So the zero function and e^{-x} have the same expansion in this region.

The next theorem gives some algebraic properties of asymptotic expansions.

Theorem C.2.2 Suppose that $f(x) \sim \Sigma a_n x^{-n}$ in D_1 and $g(x) \sim \Sigma b_n x^{-n}$ in D_2 . Then:

(i) For constants λ and μ

$$\lambda f(x) + \mu g(x) \sim \Sigma (\lambda a_n + \mu b_n) x^{-n} \text{ in } D_1 \cap D_2.$$

(ii) $f(x)g(x) \sim \Sigma c_n x^{-n}$ in $D_1 \cap D_2$, where

$$c_n = a_0 b_0 + a_1 b_{n-1} + \cdots + a_n b_0.$$

(iii) If $a_0 \neq 0$, then

$$\frac{1}{f(x)} = \sum_{k=0}^{n-1} \frac{d_k}{x^k} + O(x^{-n}) \text{ as } x \rightarrow \infty \text{ in } D_1,$$

where $a_0^{k+1} d_k$ is a polynomial in $a_0, a_1, \dots, a_k$. The d_k can be obtained from the relations

$$a_0 d_0 = 1, \quad a_0 d_k = -(a_1 d_{k-1} + a_2 d_{k-2} + \cdots + a_k d_0), \quad k = 1, 2, \dots$$

The proof of this theorem is left to the reader.

Asymptotic series can be integrated over the interval $x \leq t < \infty$, if $a_0 = a_1 = 0$.

In this case

$$f(x) \sim \frac{a_2}{x^2} + \frac{a_3}{x^3} + \cdots, \quad x \rightarrow \infty,$$

and

$$\int_x^\infty f(t) dt \sim \frac{a_2}{x} + \frac{a_3}{2x^2} + \cdots, \quad x \rightarrow \infty.$$

Integration is possible here because $f(t) = a_2 t^{-2} + O(t^{-2})$ for t large. If a_0 and a_1 are not zero, then

$$\int_x^\infty [f(t) - a_0 - a_1 t^{-1}] dt \sim \frac{a_2}{x} + \frac{a_3}{2x^3} + \cdots, \quad x \rightarrow \infty.$$

Differentiation of an asymptotic expansion may not always be valid. A standard example where differentiation fails is $f(x) = e^{-x} \sin e^x$ where $x > 0$. The derivative $f'(x) = \cos e^x - e^{-x} \sin e^x$ oscillates as $x \rightarrow \infty$ and hence does not have an asymptotic expansion by Theorem C.2.1. But

$$f(x) \sim 0 + \frac{0}{x} + \frac{0}{x^2} + \cdots, \quad x \rightarrow \infty.$$

If $f'(x)$ is continuous and has an asymptotic expansion, then it can be obtained from term-by-term differentiation of the expansion for $f(x)$. This follows from the result on integration and uniqueness of the asymptotic expansion.

C.3 Watson's Lemma

In this section, we discuss a method of obtaining the asymptotic expansion in some region of a function expressible as a Laplace integral. Section 1 has an example of an integral whose asymptotic expansion was obtained using integration by parts. The failure of this method in one case was discussed in Chapter 2. Also see Wong [1989, p. 18].

The next theorem, called Watson's lemma, gives the asymptotic expansion of $\int_0^\infty e^{-xt} f(t) dt$. See Watson [1918].

Theorem C.3.1 *Let $f(t)$ be analytic in $|t| \leq a + \delta$, where $a > 0$, $\delta > 0$, except possibly for a branch point at 0; and let*

$$f(t) = \sum_{m=1}^{\infty} a_m t^{(m/r)-1}, \quad (\text{C.3.1})$$

when $|t| \leq a$ and $r > 0$. Suppose also that $|f(t)| < \kappa e^{bt}$, where κ and b are positive numbers independent of t , when $t \geq a$. Then

$$\int_0^\infty e^{-xt} f(t) dt \sim \sum_{m=1}^{\infty} a_m \Gamma(m/r) x^{-m/r} \quad \text{as } x \rightarrow \infty \quad (\text{C.3.2})$$

for $|\arg x| \leq \frac{1}{2}\pi - \delta < \frac{1}{2}\pi$.

Proof. It is clear that for any fixed integer M , a constant C can be found such that for $t \geq 0$

$$\left| f(t) - \sum_{m=1}^{M-1} a_m t^{(m/r)-1} \right| \leq C t^{(M/r)-1} e^{bt}.$$

Hence

$$\begin{aligned} \int_0^\infty e^{-xt} f(t) dt &= \sum_{m=1}^{M-1} \int_0^\infty e^{-xt} a_m t^{(m/r)-1} dt + R_M \\ &= \sum_{m=1}^{M-1} a_m \Gamma(m/r) x^{-m/r} + R_M, \end{aligned}$$

where

$$\begin{aligned} |R_M| &\leq \int_0^\infty |e^{-xt}| C t^{(M/r)-1} e^{bt} dt \\ &= C \Gamma(M/r) / [\operatorname{Re} x - b]^{M/r}, \end{aligned}$$

provided $\operatorname{Re} x > b$. Since $|\arg x| \leq \frac{1}{2}\pi - \delta$, it follows that $\operatorname{Re} x > b$ if $|x| > b \csc \delta$. Thus,

$$|x^{M/r} R_M| < \frac{C \Gamma(M/r) |x|^{M/r}}{(|x| \sin \delta - b)^{M/r}} = O(1).$$

This proves the theorem. ■

Watson's lemma can be generalized to the case in which the integral in (C.3.2) is a contour integral, $\int_{\infty}^{(0+)} e^{-xt} f(t) dt$ and (C.3.1) is replaced by an asymptotic expansion for $f(t)$. See Olver [1974, pp. 112–115] and Wong [1989, p. 22].

C.4 The Ratio of Two Gamma Functions

We give an application of Watson's lemma to obtain the asymptotic expansion of a ratio of gamma functions. This expansion is due to Tricomi and Erdélyi [1951].

Observe that for $\operatorname{Re}(x + a) > 0$ and $\operatorname{Re}(b - a) > 0$, we have

$$\frac{\Gamma(x + a)}{\Gamma(x + b)} = \frac{1}{\Gamma(b - a)} \int_0^{\infty} e^{-xt} e^{-at} (1 - e^{-t})^{b-a-1} dt. \quad (\text{C.4.1})$$

Define the generalized Bernoulli polynomials $B_n^{\sigma}(u)$ by

$$\frac{t^{\sigma} e^{ut}}{(e^t - 1)^{\sigma}} = \sum_{n=0}^{\infty} B_n^{\sigma}(u) \frac{t^n}{n!}, \quad |t| < 2\pi. \quad (\text{C.4.2})$$

Then for $\sigma = a + 1 - b$,

$$\frac{e^{-at}}{(1 - e^{-t})^{\sigma}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} B_n^{\sigma}(a) t^{n-\sigma}, \quad |t| < 2\pi.$$

Watson's lemma now implies that

$$\begin{aligned} \frac{\Gamma(x + a)}{\Gamma(x + b)} &\sim \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} B_n^{\sigma}(a) \frac{\Gamma(b - a + n)}{\Gamma(b - a)} \frac{1}{x^{b-a+n}}, \\ |\arg x| &\leq \frac{1}{2}\pi - \delta, \quad |x| \rightarrow \infty. \end{aligned} \quad (\text{C.4.3})$$

If instead of the integral representation (C.4.1), one were to use

$$\frac{\Gamma(x + a)}{\Gamma(x + b)} = \frac{\Gamma(1 + a - b)}{2\pi i} \int_{-\infty, e^{i\alpha}}^{(0+)} e^{(x+a)t} (e^t - 1)^{b-a-1} dt, \quad (\text{C.4.4})$$

$\operatorname{Re}[(x + a)e^{i\alpha}] > 0$, $|\alpha| < \pi/2$, and for small $|t|$,

$$\alpha - \pi \leq \arg(e^t - 1) \leq \alpha + \pi;$$

then one could extend (C.4.3) to $|\arg x| \leq \pi - \delta$.

There is an improvement of (C.4.3) from the computational standpoint due to Fields [1964]. Note that if u is replaced by $\sigma - u$ in (C.4.2), then $B_k^{\sigma}(\sigma - u) = (-1)^k B_k^{\sigma}(u)$. This implies that $B_{2k+1}^{\sigma}(\sigma/2) \equiv 0$. Now write the integrand in

(C.4.1) as

$$\begin{aligned} e^{-(x+a-\sigma/2)t} e^{-\sigma t/2} (1 - e^{-t})^{-\sigma} &= e^{-(x+a-\sigma/2)t} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} B_n^{\sigma}(\sigma/2) t^{n-\sigma} \\ &= e^{-(x+a-\sigma/2)t} \sum_{n=0}^{\infty} \frac{B_{2n}^{\sigma}(\sigma/2)}{(2n)!} t^{2n-\sigma}. \end{aligned}$$

This implies

$$\frac{\Gamma(x+a)}{\Gamma(x+b)} \sim \sum_{n=0}^{\infty} \frac{B_{2n}^{\sigma}(\sigma/2)}{(2n)!} \cdot \frac{\Gamma(b-a+2n)}{\Gamma(b-a)} \frac{1}{(x+a-\sigma/2)^{2n+b-a}}$$

for $|\arg(x+a)| \leq \frac{1}{2}\pi - \delta$, $|x| \rightarrow \infty$.

Exercises

1. Let $\operatorname{Re} x \geq 0$. Show that as $x \rightarrow 0$,

$$\int_0^{\infty} \frac{e^{-t}}{1+xt} dt \sim \sum_{n=0}^{\infty} (-1)^n n! x^n$$

by two methods: (a) Expand $1/(1+xt)$ as a series. (b) Integrate by parts repeatedly.

2. Suppose that

$$F(x) = \int_0^{\infty} e^{-xt} f(t) dt$$

converges for some $x = x_0$ and f has continuous derivatives of all orders in $0 \leq t \leq a$. Show that

$$F(x) \sim \sum f^{(n)}(0) x^{-n-1}$$

uniformly in $\arg x$, as $x \rightarrow \infty$ in $|\arg x| < \pi/2 - \epsilon$ for $\epsilon > 0$.

3. Show that

$$x^{-a} e^x \int_x^{\infty} e^{-x} x^{a-1} dx \sim \frac{1}{x} + \frac{a-1}{x^2} + \frac{(a-1)(a-2)}{x^3} + \dots$$

4. Suppose θ_n is defined by

$$1 + \frac{n}{1!} + \frac{n^2}{2!} + \dots + \frac{n^{n-1}}{(n-1)!} + \frac{n^n}{n!} \theta_n = \frac{1}{2} e^n.$$

Show that

$$\theta_n = 1 + \frac{n}{2} \left(\int_0^1 (x e^{1-x})^n dx - \int_1^{\infty} (x e^{1-x})^n dx \right)$$

and hence

$$\theta_n \sim \frac{1}{3} + \frac{4}{135n} - \frac{8}{2835n^2} + \dots \quad \text{as } n \rightarrow \infty.$$

For a discussion of this result of Ramanujan, see Berndt [1989, pp. 181–184].

5. Prove Theorem C.2.2.

D

Euler–Maclaurin Summation Formula

D.1 Introduction

Some consequences of the close connection between series and integrals are brought out even in a first course in calculus. The integral test, for example, states that for a decreasing continuous function on $[1, \infty)$ the series $\sum_1^\infty f(n)$ and integral $\int_1^\infty f(x)dx$ converge or diverge together. In the derivation of Stirling's approximation for $\Gamma(x)$ given in Chapter 1, we saw that for finite sums the function f need not be decreasing for the integral to provide a good approximation. The Euler–Maclaurin summation formula makes the connection between the sum and the integral explicit for sufficiently smooth functions. In this appendix we give a statement and proof of the formula and a few applications.

Start with a differentiable function defined on a set that contains the interval $[m, n]$, where m and n are integers. Let $m \leq j < n$, where j is an integer. As a first approximation we consider

$$\int_j^{j+1} f(x)dx \approx \frac{1}{2}(f(j) + f(j+1)). \quad (\text{D.1.1})$$

It is possible to express the error in this approximation as an integral. Observe that

$$\begin{aligned} \frac{1}{2}(f(j) + f(j+1)) &= \int_j^{j+1} \frac{d}{dx} \left[\left(x - j - \frac{1}{2} \right) f(x) \right] dx \\ &= \int_j^{j+1} f(x)dx + \int_j^{j+1} \left(x - j - \frac{1}{2} \right) f'(x)dx. \end{aligned} \quad (\text{D.1.2})$$

Set $j = m, m+1, \dots, n-1$ to get

$$\frac{1}{2}(f(m) + f(m+1)) = \int_m^{m+1} f(x)dx + \int_m^{m+1} \left(x - m - \frac{1}{2} \right) f'(x)dx,$$

$$\begin{aligned}\frac{1}{2}(f(m+1) + f(m+2)) &= \int_{m+1}^{m+2} f(x)dx + \int_{m+1}^{m+2} \left(x - (m+1) - \frac{1}{2}\right) f'(x)dx, \\ \frac{1}{2}(f(n-1) + f(n)) &= \int_{n-1}^n f(x)dx + \int_{n-1}^n \left(x - (n-1) - \frac{1}{2}\right) f'(x)dx.\end{aligned}$$

The second integrals on the right can be added together if we note that the integrand can be written as $(x - [x] - \frac{1}{2})f'(x)$. Addition gives

$$\sum_{m+1}^{n-1} f(k) + \frac{1}{2}(f(m) + f(n)) = \int_m^n f(x)dx + \int_m^n \left(x - [x] - \frac{1}{2}\right) f'(x)dx. \quad (\text{D.1.3})$$

This is a particular case of the Euler–Maclaurin formula. To put it in a slightly different form, set $B_1(x) = x - \frac{1}{2}$, so that $B_1(x - [x]) = x - [x] - \frac{1}{2}$. Also let $B_1 = B_1(0) = -1/2$. Now write (D.1.3) as

$$\sum_{m+1}^n f(k) + B_1\{f(n) - f(m)\} = \int_m^n f(x)dx + \int_m^n B_1(x - [x])f'(x)dx. \quad (\text{D.1.4})$$

Recall that $B_1(x)$ is the first Bernoulli polynomial and B_1 the first Bernoulli number.

As an application take $f(x) = \log x$ and $m = 1$ in (D.1.4) to get

$$\log n! - \left(n + \frac{1}{2}\right) \log n + n = 1 + \int_1^n \frac{B_1(x - [x])}{x} dx. \quad (\text{D.1.5})$$

Since $B_1(x - [x])$ is periodic of period 1 and

$$\int_0^1 B_1(x - [x])dx = \int_t^{t+1} B_1(x - [x])dx = 0 \quad (\text{D.1.6})$$

for any $t \geq 0$, we see that the limit as $n \rightarrow \infty$ in (D.1.5) exists and is equal to

$$1 + \int_1^\infty \frac{B_1(x - [x])}{x} dx. \quad (\text{D.1.7})$$

This is exactly Stirling's approximation, if we can show that the expression (D.1.7) equals $\frac{1}{2} \log 2\pi$. We prove this later by a method different from the one given in Chapter 1, which uses Wallis's formula.

Formula (D.1.4) was obtained by one integration by parts. With Stirling's formula in mind, it makes sense to repeat the process. This would give an expression with higher derivatives of $f(x)$. If $f(x) = \log x$, then $f'(x) = 1/x$, $f''(x) = -1/x^2$, and so on. These quantities get small for large x and so an extension of (D.1.4) would be useful.

D.2 The Euler–Maclaurin Formula

Let $\tilde{B}_2(x)$ be a primitive (antiderivative) of $B_1(x - [x]) = \tilde{B}_1(x)$. It follows from (D.1.6) that $\tilde{B}_2$ is periodic with period one. In particular, this implies that $\tilde{B}_2(0) = \tilde{B}_2(1) = \cdots = \tilde{B}_2(j) = \tilde{B}_2(j+1) = \cdots$. Now, integration by parts gives

$$\int_j^{j+1} \tilde{B}_1(x) f'(x) dx = \tilde{B}_2(0)(f'(j+1) - f'(j)) - \int_j^{j+1} \tilde{B}_2(x) f''(x) dx.$$

We assume that f has continuous derivatives of as high an order as necessary.

Sum from $j = m$ to $j = n$ to get

$$\int_m^n \tilde{B}_1(x) f'(x) dx = \tilde{B}_2(0)(f'(n) - f'(m)) - \int_m^n \tilde{B}_2(x) f''(x) dx. \quad (\text{D.2.1})$$

Note that it was the periodicity of $\tilde{B}_2(x)$ that gave us the simple expression on the right side of Equation (D.2.1). This suggests that we should choose the constant of integration in $\tilde{B}_2(x)$ such that

$$\int_0^1 \tilde{B}_2(x) dx = \int_t^{t+1} \tilde{B}_2(x) dx = 0. \quad (\text{D.2.2})$$

Let $\frac{1}{2}B_2(x) = \tilde{B}_2(x)$ for $0 \leq x \leq 1$, so that $\tilde{B}_2(x) = \frac{1}{2}B_2(x - [x])$. From the definition of $\tilde{B}_2(x)$, it follows that

$$B_2(x) = x^2 - x + \frac{1}{6},$$

which is the second Bernoulli polynomial. We state the general Euler–Maclaurin formula in the next theorem:

Theorem D.2.1 *Suppose f has continuous derivatives up to order s . Then*

$$\begin{aligned} \sum_{m+1}^n f(x) &= \int_m^n f(x) dx + \sum_{\ell=1}^s (-1)^\ell \frac{B_\ell}{\ell!} \{f^{(\ell-1)}(n) - f^{(\ell-1)}(m)\} \\ &\quad + \frac{(-1)^{s-1}}{s!} \int_m^n B_s(x - [x]) f^{(s)}(x) dx, \end{aligned}$$

where $B_s(x)$ are the Bernoulli polynomials and $B_\ell = B_\ell(0)$, the Bernoulli numbers.

Proof. As in the derivation of (D.2.1), apply integration by parts successively to obtain a sequence of periodic functions $\tilde{B}_n(x)$ such that $\tilde{B}_n'(x) = \tilde{B}_{n-1}(x)$ ($0 < x < 1$, $n \geq 1$) and $\int_0^1 \tilde{B}_n(x) dx = 0$, $n \geq 1$. With respect to these functions,

we obtain the formula

$$\begin{aligned} \sum_{m+1}^n f(k) &= \int_m^n f(x)dx + \sum_{\ell=1}^s (-1)^\ell \tilde{B}_\ell(0) \{f^{(\ell-1)}(n) - f^{(\ell-1)}(m)\} \\ &\quad + (-1)^{s-1} \int_m^n \tilde{B}_s(x) f^{(s)}(x)dx. \end{aligned}$$

To show the relation of $\tilde{B}_\ell(x)$ to Bernoulli polynomials, consider the generating function

$$G(x, t) := \sum_0^\infty \tilde{B}_n(x) t^n, \quad \text{where } \tilde{B}_0(x) = 1, \quad (\text{D.2.3})$$

of the sequence $\{\tilde{B}_\ell(x)\}$. Observe that for $0 < x < 1$,

$$\frac{\partial G}{\partial x} = \sum_{n=1}^\infty \tilde{B}'_n(x) t^n = \sum_{n=1}^\infty \tilde{B}_{n-1}(x) t^n = t \sum_{n=0}^\infty \tilde{B}_n(x) t^n = tG.$$

We then have

$$G(x, t) = A(t)e^{xt}. \quad (\text{D.2.4})$$

Use $\int_0^1 \tilde{B}_n(x)dx = 0$ in (D.2.3) and (D.2.4) to obtain

$$1 = A(t) \frac{e^t - 1}{t} \quad \text{or} \quad A(t) = \frac{t}{e^t - 1}.$$

Thus

$$G(x, t) = \frac{te^{xt}}{e^t - 1} = \sum_0^\infty B_n(x) \frac{t^n}{n!}, \quad (\text{D.2.5})$$

and

$$\tilde{B}_n(x) = \frac{B_n(x)}{n!} \quad \text{for } 0 < x < 1.$$

By the periodicity of $\tilde{B}_n(x)$ we arrive at

$$\tilde{B}_n(x) = \frac{1}{n!} B_n(x - [x]) \quad \text{for all } x.$$

Although this formal argument has not been justified, it can be done easily since the generating function (D.2.5) is analytic in the disk $|t| < 2\pi$. This proves the theorem. ■

D.3 Applications

Take $f(x) = x^{-s} \log x$, $m = 1$, and $n = N$ in (D.1.3) to get

$$\begin{aligned} \sum_{n=2}^N \frac{\log n}{n^s} &= \frac{N^{1-s} \log N}{1-s} + \frac{1}{(s-1)^2} - \frac{N^{1-s}}{(s-1)^2} \\ &\quad + \frac{1}{2}(N^{-s} \log N) + \int_1^N \frac{x - [x] - \frac{1}{2}}{x^{s+1}} (1-s \log x) dx. \end{aligned}$$

Let $\operatorname{Re} s > 1$ and $N \rightarrow \infty$ to obtain

$$\zeta'(s) = \frac{-1}{(s-1)^2} - \int_1^\infty \frac{x - [x] - \frac{1}{2}}{x^{s+1}} (1-s \log x) dx.$$

The integral on the right side of the equality converges for $\operatorname{Re} s > -1$ by Abel's test. So

$$\zeta'(0) = -1 - \int_1^\infty \frac{x - [x] - \frac{1}{2}}{x} dx. \quad (\text{D.3.1})$$

By Corollary 1.3.3 we have $\zeta'(0) = -\frac{1}{2} \log 2\pi$. Thus the constant (D.1.7) in Stirling's formula is $\frac{1}{2} \log 2\pi$.

As another application, we prove the following useful theorem on the order of $\zeta(s)$ and $\zeta'(s)$ for large $\operatorname{Im} s$.

Theorem D.3.1 *Let $s = \sigma + it$. For $1 - \frac{A}{\log |t|} \leq \sigma \leq 2$, where A is any positive constant, and $|t|$ large enough,*

$$\zeta(s) = O(\log |t|)$$

and

$$\zeta'(s) = O(\log^2 |t|).$$

Proof. In (D.1.3), set $f(x) = x^{-s}$, $m = N$, and let $n \rightarrow \infty$ to arrive at

$$\zeta(s) = \sum_{n=1}^N \frac{1}{n^s} + \frac{N^{1-s}}{s-1} - \frac{1}{2} N^{-s} - s \int_N^\infty \frac{x - [x] - \frac{1}{2}}{x^{s+1}} dx.$$

The integral is $O(|t|/\sigma N^\sigma)$. Note that

$$|n^{-s}| = n^{-\sigma} \leq n^{-(1-A/\log |t|)} = O(1/n),$$

where the last equality holds for $n \leq |t|$. Thus, take $N = [|t|]$ to get

$$\zeta(s) = \sum_{n=1}^N O\left(\frac{1}{n}\right) + O(1) = O(\log |t|).$$

To derive the result for $\zeta'(s)$, take $f(x) = x^{-s} \log x$ and do a similar calculation. See Rademacher [1973, p. 100], if necessary. ■

We complete this section with a proof of the following theorem stated in Chapter 1.

Theorem D.3.2 *Let $z \in \mathbb{C} - (-\infty, 0]$. Then*

$$\begin{aligned} \log \Gamma(z) = & \left(z - \frac{1}{2}\right) \log z - z + \sum_{j=1}^m \frac{B_{2j}}{2j(2j-1)} \frac{1}{z^{2j-1}} \\ & + \frac{1}{2} \log 2\pi - \frac{1}{2m} \int_0^\infty \frac{B_{2m}(x - [x])}{(x+z)^{2m}} dx. \end{aligned}$$

Proof. Start with the following expression of the gamma function:

$$\Gamma(z) = \lim_{n \rightarrow \infty} \prod_{\ell=1}^n \frac{\ell}{z + \ell - 1} \left(\frac{\ell+1}{\ell}\right)^{z-1}.$$

Then

$$\log \Gamma(z) = \lim_{n \rightarrow \infty} \left[(z-1) \log(n+1) - \sum_{\ell=1}^n \log \frac{z + \ell - 1}{\ell} \right], \quad (\text{D.3.2})$$

where the principal branch of the log function in $\mathbb{C} - (-\infty, 0]$ is chosen. In Theorem D.2.1, take

$$f(x) = \log \frac{x+z-1}{x} = \log(x+z-1) - \log x$$

to get

$$\begin{aligned} \sum_{\ell=1}^n \log \frac{z + \ell - 1}{\ell} &= \log z + \sum_{\ell=2}^n \log \frac{z + \ell - 1}{\ell} \\ &= \log z + \int_1^n [\log(x+z-1) - \log x] dx \\ &\quad + \sum_{j=1}^m \frac{B_{2j}}{2j(2j-1)} \left[\frac{1}{(n+z-1)^{2j-1}} - \frac{1}{n^{2j-1}} - \frac{1}{z^{2j-1}} + 1 \right] \\ &\quad + \frac{1}{2} [\log(n+z-1) - \log n - \log z] \\ &\quad + \frac{1}{2m} \int_0^n B_{2m}(x - [x]) \left[\frac{1}{(z+x-1)^{2m}} - \frac{1}{x^{2m}} \right] dx. \end{aligned}$$

Here we have used the fact that $B_1 = -1/2$ and $B_{2j+1} = 0$ for $j \geq 1$. Compute the first of the above integrals and observe that after some cancellation the terms

that involve n are

$$(n+z-1) \log(n+z-1) - n \log n + \frac{1}{2} \log \frac{n+z-1}{n} \\ + \sum_{j=1}^m \frac{B_{2j}}{2j(2j-1)} \left[\frac{1}{(n+z-1)^{2j-1}} - \frac{1}{n^{2j-1}} \right].$$

Subtract this from $(z-1) \log(n+1)$ and let $n \rightarrow \infty$ to compute the limit in (D.3.2). The result is

$$\log \Gamma(z) = \left(z - \frac{1}{2} \right) \log z - z + 1 + \sum_{j=1}^m \frac{B_{2j}}{2j(2j-1)} \left(\frac{1}{z^{2j-1}} - 1 \right) \\ - \frac{1}{2m} \int_1^\infty B_{2m}(x - [x]) \left[\frac{1}{(x+z-1)^{2m}} - \frac{1}{x^{2m}} \right] dx. \quad (\text{D.3.3})$$

From (D.1.5) and (D.3.1) we know that

$$\lim_{z \rightarrow \infty} [\log \Gamma(z) - (z - 1/2) \log z + z] = \frac{1}{2} \log 2\pi.$$

So let $z \rightarrow \infty$ in (D.3.3) to see that

$$1 - \sum_{j=1}^m \frac{B_{2j}}{2j(2j-1)} + \frac{1}{2m} \int_1^\infty \frac{B_{2m}(x - [x])}{x^{2m}} dx = \frac{1}{2} \log 2\pi.$$

This result combined with (D.3.3) gives the formula in Theorem D.3.2. ■

D.4 The Poisson Summation Formula

In this section we state and prove an important and useful formula from the theory of Fourier series. This is the Poisson summation formula. It has numerous applications though we mention only a few. One consequence of Poisson's formula is the Euler-Maclaurin summation formula.

Start with a result on Fourier series attributed to Jordan. Recall that a function $f(x)$ is said to be of bounded variation on an interval $[a, b]$ (which may be infinite), if there is a constant $C > 0$ such that for any set of points $x_0 < x_1 < \cdots < x_n$ in the interval

$$\sum_{k=1}^n |f(x_k) - f(x_{k-1})| < C.$$

If the interval is the whole real line, then we say that f is of bounded total variation. An important though easily proved result is that if f is of bounded variation on $[a, b]$, then $f = g - h$, where g and h are increasing functions on $[a, b]$.

Jordan's theorem is the following: Suppose f is integrable on $[0, 1]$ and periodic with period one. If f is of bounded variation on $[0, 1]$, then

$$\frac{f(x+) + f(x-)}{2} = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \left(\int_0^1 f(t) e^{-2\pi i n t} dt \right) e^{2\pi i n x}.$$

Poisson's summation formula is contained in the next theorem.

Theorem D.4.1 Suppose g is integrable on $(-\infty, \infty)$ and of bounded total variation. Suppose h is an even, positive, integrable function that is decreasing on $[0, \infty)$ and such that $|g(x)| \leq h(x)$. Then

$$\sum_{n=-\infty}^{\infty} \frac{g(x+n+) + g(x+n-)}{2} = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \int_{-\infty}^{\infty} g(t) e^{-2\pi i n t} dt e^{2\pi i n x}. \quad (\text{D.4.1})$$

Proof. The inequality

$$\sum_{n=-M}^{N-1} |g(x+n)| \leq |g(x)| + \int_{-M}^N h(x+t) dt$$

implies that $\sum_{n=-\infty}^{\infty} g(x+n)$ is an absolutely convergent series and defines a periodic function $f(x)$ with period one. Moreover, since g is of bounded total variation, f is of bounded variation on $[0, 1]$. By Jordan's theorem

$$\frac{f(x+) + f(x-)}{2} = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \int_0^1 f(t) e^{-2\pi i n t} dt e^{2\pi i n x}.$$

Now

$$\begin{aligned} \int_0^1 f(t) e^{-2\pi i n t} dt &= \int_0^1 \sum_{-\infty}^{\infty} g(m+t) e^{-2\pi i n t} dt \\ &= \sum_{-\infty}^{\infty} \int_0^1 g(m+t) e^{-2\pi i n t} dt \\ &= \sum_{-\infty}^{\infty} \int_m^{m+1} g(u) e^{-2\pi i n u} du \\ &= \int_{-\infty}^{\infty} g(u) e^{-2\pi i n u} du. \end{aligned}$$

The summation in the second line can be taken outside the integral by the dominated convergence theorem. ■

Remark D.4.1 If the series on the right-hand side in Theorem D.4.1 is absolutely convergent, then we can write

$$\sum_{n=-\infty}^{\infty} \frac{g(x+n+) + g(x+n-)}{2} = \sum_{n=-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(t) e^{-2\pi i n t} dt \right) e^{2\pi i n x}.$$

An interesting example of Poisson's formula connects the Poisson kernel for the upper half plane given by $g(x) = y/(x^2 + y^2)$, $y > 0$, with the kernel for the unit disk given by $(1 - r^2)/(1 - 2r \cos x + r^2)$. Observe that with g as defined here,

$$\hat{g}(t) = \int_{-\infty}^{\infty} \frac{y e^{-2\pi i t x}}{x^2 + y^2} dx = \pi e^{-2\pi y|t|}.$$

The reader may verify this by contour integration or by other means. Substitution of this in Theorem D.4.1 gives

$$\sum_{n=-\infty}^{\infty} \frac{y}{y^2 + (x+n)^2} = \sum_{n=-\infty}^{\infty} e^{-2\pi y|n|} e^{2\pi i n x} = \frac{1 - e^{-4\pi y}}{1 - 2e^{-2\pi y} \cos 2\pi x + e^{-4\pi y}}.$$

The partial fraction expansions for the trigonometric functions can also be derived from Theorem D.4.1. Take $s > 0$ and set

$$g(t) = \begin{cases} e^{-st}, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

The left side of Poisson's formula becomes

$$\sum_{n > -x} e^{-s(n+x)} + \begin{cases} 1/2, & x \text{ integer,} \\ 0, & \text{otherwise.} \end{cases}$$

The right-hand side is given by

$$\sum_{n=-\infty}^{\infty} \frac{e^{2\pi i n x}}{s + 2\pi i n}.$$

When $x = 0$, we get

$$\frac{e^s + 1}{e^s - 1} = \frac{1}{s} + \sum_{n=1}^{\infty} \left(\frac{2}{s + 2\pi i n} + \frac{1}{s - 2\pi i n} \right).$$

By analytic continuation, we can take $s = 2\pi i y$. The result is

$$\pi \cot \pi y = \frac{1}{y} + \sum_{n=1}^{\infty} \left(\frac{1}{y+n} + \frac{1}{y-n} \right).$$

When $x = 1/2$, we obtain

$$\frac{e^{s/2}}{e^s - 1} = \frac{1}{s} + \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{s + 2\pi in} + \frac{(-1)^n}{s - 2\pi in} \right).$$

This time, $s = 2\pi iy$ gives

$$\pi \csc \pi y = \frac{1}{y} + \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{y + n} + \frac{1}{y - n} \right).$$

An important transformation formula for theta functions follows from the Poisson summation formula by taking $g(x) = e^{-s\pi x^2}$. The formula is

$$\sum_{n=-\infty}^{\infty} e^{-s\pi(n+x)^2} = s^{-1/2} \sum_{n=-\infty}^{\infty} e^{-\pi n^2/s} e^{2\pi inx}. \quad (\text{D.4.2})$$

The Euler–Maclaurin formula follows from the Poisson summation formula by taking

$$g(x) = \begin{cases} f(x), & \text{for } a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

Here a and b are assumed to be nonnegative integers and $f(x)$ is continuously differentiable in $[a, b]$. This implies that f is of bounded variation on $[a, b]$ and hence $g(x)$ is of bounded total variation. Use this $g(x)$ in (D.4.1). The result is

$$\begin{aligned} \frac{1}{2}(f(a) + f(b)) + \sum_{k=a+1}^{b-1} f(k) &= \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \int_a^b f(t) e^{-2\pi int} dt \\ &= \int_a^b f(t) dt + 2 \sum_{n=1}^{\infty} \int_a^b f(t) \cos 2\pi n t dt. \end{aligned} \quad (\text{D.4.3})$$

Integration by parts gives

$$\int_a^b f(t) \cos 2\pi n t dt = - \int_a^b f'(t) \frac{\sin 2\pi n t}{2n\pi} dt.$$

Also,

$$2 \sum_{n=1}^{\infty} \frac{\sin 2\pi n x}{2n\pi} = -B_1(x - [x]),$$

except when x is an integer (see Exercise 44 in Chapter 1). Thus, by the dominated convergence theorem, we can write (D.4.2) as

$$\sum_{a < n \leq b} f(n) = \int_a^b f(t) dt - B_1(f(b) - f(a)) + \int_a^b f'(x) B_1(x - [x]) dx.$$

If we assume that $f^{(m)}(x)$ is continuously differentiable, then the application of

$$\frac{dB_n(x)}{dx} = nB_{n-1}(x), \quad 0 < x < 1,$$

and successive integration by parts gives Theorem D.2.1. For other applications and extensions of the summation formulas studied here, see Berndt [1975].

Exercises

1. Prove that with the notation and conditions of Theorem D.3.1 $\zeta'(s) = O(\log^2 |t|)$.
2. Define $\Delta f(0) = f(1) - f(0)$. Show that if $f(x)$ is a polynomial of degree m , then

$$f(0) = \int_0^1 f(x)dx + \sum_{j=1}^m \frac{B_j}{j} \Delta f^{(j-1)}(0).$$

3. Use the Euler–Maclaurin formula to obtain the relation

$$\sum_{k=1}^n \frac{1}{k} = \log n + \gamma + \frac{1}{2n} - \sum_{k=1}^{p-1} \frac{B_{2k}}{2kn^{2k}} - \frac{\theta B_{2p}}{2pn^{2p}},$$

where $0 < \theta < 1$ and γ is Euler's constant.

4. Show that there is a constant C such that

$$\sum_{2 \leq n \leq x} \frac{1}{n \log n} = \log \log x + C + O\left(\frac{1}{x \log x}\right).$$

5. Use the Poisson summation formula to prove that

$$\sum_{-\infty}^{\infty} e^{-s(n+x)^2} = \frac{1}{\sqrt{s}} \sum_{-\infty}^{\infty} e^{-\pi n^2/s} e^{2\pi i n x}.$$

6. Use Poisson summation to obtain the following theorem of Lipschitz: Suppose $\operatorname{Re} x > 0$, $0 < \alpha \leq 1$, and $\operatorname{Re} s > 1$. Then

$$\frac{(2\pi)^s}{\Gamma(s)} \sum_{n=0}^{\infty} (n + \alpha)^{s-1} e^{-2\pi x(n+\alpha)} = \sum_{n=-\infty}^{\infty} \frac{e^{2\pi i n \alpha}}{(x + ni)^s}.$$

(For a different proof where $\alpha = 1$, see Exercise 37 in Chapter 2.)

7. Suppose a , b , and c are integers such that $ac + b$ is even. Use Poisson summation to prove the reciprocity for Gaussian sums

$$\sum_{n=0}^{|c|-1} e^{\pi i (an^2 + bn)/c} = \sqrt{\left|\frac{c}{a}\right|} e^{\pi i (|ac| - b^2)/(4ac)} \sum_{n=0}^{|a|-1} e^{-\pi i (cn^2 + bn)/a}.$$

E

Lagrange Inversion Formula

E.1 Reversion of Series

There are situations in analysis in which one knows a series expansion for $y(x)$ but would like to obtain the series for x in terms of y . Newton, for example, encountered such a problem when he had the series for $\sin^{-1} x$ by integrating the expansion of $(1 - x^2)^{-1/2}$ term by term and he wanted the series for $\sin x$. Some results of Newton on reversion of series can be found in Newton [1960, p. 147].

Suppose a series for x in powers of y is required when $x = y\phi(x)$. Assume that ϕ is analytic in a neighborhood of $x = 0$ with $\phi(0) \neq 0$. Then

$$y = x/\phi(x) = \sum_{n=1}^{\infty} a_n x^n, \quad a_1 \neq 0. \quad (\text{E.1.1})$$

We shall see below that Lagrange's inversion formula gives

$$x = \sum_{n=1}^{\infty} b_n y^n,$$

where

$$b_n = \frac{1}{n!} \frac{d^{n-1}}{dx^{n-1}} [\phi(x)^n]_{x=0}.$$

This means that nb_n is the coefficient of x^{n-1} in the expansion of $\phi^n(x)$ or the coefficient of x^{-1} in the expansion of $1/y^n$.

More generally, suppose that (E.1.1) holds and that $f(x)$ is an analytic function in a neighborhood of $x = 0$. Then Lagrange's formula is

$$f(x) = f(0) + \sum_{n=1}^{\infty} \frac{y^n}{n!} \left[\frac{d^{n-1}}{dx^{n-1}} (f'(x)\phi^n(x)) \right]_{x=0}. \quad (\text{E.1.2})$$

E.2 A Basic Lemma

The proof of Lagrange's formula (E.1.2) depends on a simple lemma given below. The treatment here follows that of Gessel and Stanton [1982], which is based on the work of Jacobi [1830]. See also Bromwich [1926]. An extension of Lagrange inversion to several variables is in Good [1960] and q -analogs are discussed in Stanton [1988]. A history of this formula can be found in W. Johnson's unpublished manuscript, *Notes on the Lagrange inversion formula*.

Define the residue of a formal Laurent series $f(x) = \sum_{j=-m}^{\infty} a_j x^j$ by $\text{Res}[f(x)] = a_{-1}$. The next lemma states that the residue does not change under a certain change of variables.

Lemma E.2.1 Let $G(x) = \sum_{j=-m}^{\infty} a_j x^j$ and $h(x) = \sum_{i=1}^{\infty} b_i x^i$, where $b_1 \neq 0$. Then $\text{Res}[G(h(x))h'(x)] = \text{Res}[G(x)]$.

Proof. Since both sides of the equation are linear in G , it is sufficient to prove it for $G(x) = x^m$. Since $\text{Res}[g'(x)] = 0$ for any Laurent series $g(x)$, it follows that, for $m \neq -1$,

$$\text{Res}[h^m(x)h'(x)] = \frac{1}{m+1} \text{Res}[\{h^{m+1}(x)\}'] = 0.$$

For $m = -1$, let $h(x) = b_1 x f(x)$. Then

$$\text{Res}\left[\frac{h'(x)}{h(x)}\right] = \text{Res}\left[\frac{1}{x} + \frac{f'(x)}{f(x)}\right] = 1 + \text{Res}[\{\log f(x)\}'] = 1.$$

The lemma is proved. ■

To prove (E.1.2) note that $y = a_1 x + a_2 x^2 + \cdots$ (x a complex variable) is conformal in a neighborhood of 0, because $a_1 \neq 0$, and so x can be expanded in powers of y . Since $f(x)$ is analytic, it can be expanded in powers of y . Suppose that

$$f(x) = f(0) + \sum_{n=1}^{\infty} c_n y^n.$$

Then

$$f'(x) = \sum_{n=1}^{\infty} n c_n y^{n-1} \frac{dy}{dx}.$$

Let $G(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$. By Lemma E.2.1,

$$\text{Res}\left[\frac{f'(x)\phi'(x)}{x^r}\right] = \text{Res}\left[\frac{f'(x)}{y^r}\right] = \text{Res}\left[\frac{G(y)dy/dx}{y^r}\right] = \text{Res}\left[\frac{G(x)}{x^r}\right] = r c_r.$$

But the Taylor series of $f'(x)\phi^r(x)$ is

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} \left[\frac{d^n}{dx^n} (f'(x)\phi^r(x)) \right]_{x=0},$$

and so

$$\text{Res} \left[\frac{f'(x)\phi^r(x)}{x^r} \right] = \frac{1}{(r-1)!} \left[\frac{d^{r-1}}{dx^{r-1}} (f'(x)\phi^r(x)) \right]_{x=0}.$$

Thus

$$c_r = \frac{1}{r!} \left[\frac{d^{r-1}}{dx^{r-1}} (f'(x)\phi^r(x)) \right]_{x=0},$$

which proves Lagrange's formula (E.1.2).

A variant of (E.1.2) is the formula

$$\frac{f(x)}{1 - y\phi(x)} = \sum_{n=0}^{\infty} \frac{y^n}{n!} \left[\frac{d^n}{dx^n} (f(x)\phi^n(x)) \right]_{x=0}, \quad (\text{E.2.1})$$

where y is defined by (E.1.1). To prove (E.2.1), note that since $x - y\phi(x) = 0$, we have

$$1 - \phi(x) \frac{dy}{dx} - y\phi'(x) = 0,$$

or

$$\frac{dy}{dx} = \frac{1 - y\phi'(x)}{\phi(x)}. \quad (\text{E.2.2})$$

Take the derivative of (E.1.2) with respect to x and use (E.2.2) to get

$$\frac{f'(x)\phi(x)}{1 - y\phi'(x)} = \sum_{n=0}^{\infty} \frac{y^n}{n!} \left[\frac{d^n}{dx^n} (f'(x)\phi(x)\phi^n(x)) \right]_{x=0}.$$

Set $F(x) = f'(x)\phi(x)$ to obtain (E.2.1).

E.3 Lambert's Identity

Lagrange's inversion formula can be used to obtain a number of interesting series expansions for specific functions. A few examples are given in the exercises. Here we derive an identity of Lambert. This is given by

$$(1+x)^\alpha = 1 + \sum_{n=1}^{\infty} \frac{\alpha}{\alpha + n\beta} \binom{\alpha + \beta n}{n} y^n, \quad (\text{E.3.1})$$

where $y = x(1+x)^{-\beta}$. Take $f(x) = (1+x)^\alpha$ and $\phi(x) = (1+x)^\beta$ in (E.1.2). Then

$$(1+x)^\alpha = 1 + \sum_{n=1}^{\infty} \frac{y^n}{n!} \left[\frac{d^{n-1}}{dx^{n-1}} (\alpha(1+x)^{n\beta+\alpha-1}) \right]_{x=0}.$$

The expression in square brackets equals

$$\alpha(n\beta + \alpha - 1) \cdots (n\beta + \alpha - n + 1) = \alpha(n-1)! \binom{\alpha + n\beta - 1}{n-1},$$

and this gives (E.3.1). The identity corresponding to (E.2.1) is

$$\frac{(1+x)^{\alpha+1}}{1+x(1-\beta)} = 1 + \sum_{n=1}^{\infty} \binom{\alpha + \beta n}{n} y^n. \quad (\text{E.3.2})$$

An interesting binomial identity comes from (E.3.1) and (E.3.2). Equate the coefficient of y^n on both sides of

$$(1+x)^\gamma \frac{(1+x)^{\alpha+1}}{1+x(1-\beta)} = \frac{(1+x)^{\alpha+\gamma+1}}{1+x(1-\beta)}.$$

The result is

$$\sum_{k=0}^n \frac{\gamma}{\gamma + \beta k} \binom{\gamma + \beta k}{k} \binom{\alpha + \beta(n-k)}{n-k} = \binom{\alpha + \gamma + \beta n}{n}. \quad (\text{E.3.3})$$

E.4 Whipple's Transformation

Gessel and Stanton showed how to derive Whipple's transformation, which we obtained in different ways in Chapters 3 and 7, from Lemma E.2.1.

Theorem E.4.1 Suppose $A(x)$, $B(x)$, $C(x)$, and $D(x)$ are power series whose coefficients of x^k are A_k , B_k , C_k , and D_k respectively. Suppose

$$(1-x)^{-\alpha} A(x/(1-x)^{\beta+1}) = B(x), \quad (\text{E.4.1})$$

$$(1+\beta x)(1-x)^{-\gamma} C(x/(1-x)^{\beta+1}) = D(x). \quad (\text{E.4.2})$$

If $n(\beta+1) = 1 - \alpha - \gamma$, then

$$\sum_{k=0}^n B_k D_{n-k} = \sum_{k=0}^n A_k C_{n-k}.$$

Proof. First note that

$$\text{Res} \left[\frac{B(x)D(x)}{x^{n+1}} \right] = \text{coefficient of } x^n \text{ in } B(x)D(x) = \sum_{k=0}^n B_k D_{n-k}.$$

Now write $h(x) = x(1-x)^{-\beta-1}$ so that $h'(x) = (1+\beta x)(1-x)^{-\beta-2}$ and

$$\begin{aligned} \frac{B(x)D(x)}{x^{n+1}} &= \frac{(1+\beta x)(1-x)^{-\alpha-\gamma} A(x/(1-x)^{\beta+1}) C(x/(1-x)^{\beta+1})}{x^{n+1}} \\ &= \frac{(1+\beta x)(1-x)^{n(\beta+1)-1} A(h(x)) C(h(x))}{x^{n+1}} \\ &= \frac{A(h(x)) C(h(x)) h'(x)}{[h(x)]^{n+1}}. \end{aligned}$$

By Lemma E.2.1,

$$\operatorname{Res} \left[\frac{B(x)D(x)}{x^{n+1}} \right] = \operatorname{Res} \left[\frac{A(x)C(x)}{x^{n+1}} \right].$$

By the remark at the beginning of the proof, this implies that

$$\sum_{k=0}^n B_k D_{n-k} = \sum_{k=0}^n A_k C_{n-k}.$$

This proves the theorem. ■

To derive Whipple's transformation for a ${}_7F_6$, recall Whipple's quadratic transformation noted in Chapter 2, namely,

$$\begin{aligned} (1-x)^{-a} {}_3F_2 \left(\begin{matrix} a/2, (1+a)/2, 1+a-b-c \\ 1+a-b, 1+a-c \end{matrix} ; -\frac{4x}{(1-x)^2} \right) \\ = {}_3F_2 \left(\begin{matrix} a, b, c \\ 1+a-b, 1+a-c \end{matrix} ; x \right). \end{aligned} \quad (\text{E.4.3})$$

This corresponds to (E.4.1) with $\alpha = a$ and $\beta = 1$. The transformation corresponding to (E.4.3) is a result of Bailey [1929]:

$$\begin{aligned} (1+x)(1-x)^{-a-1} {}_3F_2 \left[\begin{matrix} (a+1)/2, 1+a/2, a+1-b-c \\ 1+a-b, 1+a-c \end{matrix} ; -\frac{4x}{(1-x)^2} \right] \\ = {}_4F_3 \left[\begin{matrix} a, 1+a/2, b, c \\ a/2, 1+a-b, 1+a-c \end{matrix} ; x \right]. \end{aligned} \quad (\text{E.4.4})$$

This can be obtained from (E.4.3) by differentiation or directly by equating the coefficients of x^n . To apply Theorem E.4.1, note that from (E.4.3),

$$\begin{aligned} A_k &= (a/2)_k ((1+a)/2)_k (1+a-b-c)_k (-4)^k / [k! (1+a-b)_k (1+a-c)_k], \\ B_k &= (a)_k (b)_k (c)_k / [k! (1+a-b)_k (1+a-c)_k], \end{aligned}$$

and from (E.4.4) after renaming the parameters,

$$\begin{aligned} C_k &= ((1+d)/2)_k ((2+d)/2)_k (1+d-e-f)_k (-4)^k / \\ &\quad [k! (1+d-e)_k (1+d-f)_k], \\ D_k &= (d)_k (1+d/2)_k (e)_k (f)_k / [k! (d/2)_k (1+d-e)_k (1+d-f)_k]. \end{aligned}$$

So

$$\begin{aligned} \sum_{k=0}^n A_k C_{n-k} \\ = \frac{(-4)^n ((1+d)/2)_n ((2+d)/2)_n (1+d-e-f)_n}{n! (1+d-e)_n (1+d-f)_n} \\ \cdot \sum_{k=0}^n \frac{(a/2)_k ((1+a)/2)_k (1+a-b-c)_k (-n)_k (e-d-n)_k (f-d-n)_k}{k! (1+a-b)_k (1+a-c)_k ((1-d)/2-n)_k (-n-d/2)_k (e+f-n-d)_k}. \end{aligned}$$

Theorem E.4.1 is applicable when $2n = -a - d$. This simplifies the expression in the sum. For example, $-n - d/2 = a/2$, and the terms involving these expressions cancel. After simplification the sum becomes

$${}_4F_3\left(\begin{matrix} -n, 1+a-b-c, e+a+n, f+a+n \\ 1+a-b, 1+a-c, e+f+a+n \end{matrix}; 1\right).$$

The sum $\sum_{k=0}^n B_k D_{n-k}$ leads to a ${}_7F_6$ and the equality

$$\sum_{k=0}^n B_k D_{n-k} = \sum_{k=0}^n A_k C_{n-k}$$

results in Whipple's theorem

$$\begin{aligned} & {}_7F_6\left(\begin{matrix} a, 1+a/2, b, c, e+a+n, f+a+n, -n \\ a/2, 1+a-b, 1+a-c, 1-e-n, 1-f-n, 1+a+n \end{matrix}; 1\right) \\ &= \frac{(a+1)_n(a+c+f+n)_n}{(e)_n(f)_n} \\ &\quad \times {}_4F_3\left(\begin{matrix} 1+a-b-c, e+a+n, f+a+n, -n \\ 1+a-b, 1+a-c, e+f+a+n \end{matrix}; 1\right). \end{aligned}$$

Exercises

1. Show that

$$e^{\alpha x} = \sum_{n=0}^{\infty} \frac{\alpha(\alpha + bn)^{n-1}}{n!} (xe^{-bx})^n$$

and that

$$x = \sum_{n=1}^{\infty} \frac{n^{n-1}}{n!} (xe^{-x})^n.$$

2. Show formally that

$$f(x+a) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a-nb)}{n!} x(x+nb)^{n-1}.$$

Give sufficient conditions for this to be correct.

3. Show that

$$(a+x)^\alpha = \sum_{k=0}^{\infty} x(x-kb)^{k-1} \frac{(-\alpha)_k}{k!} (-1)^k (a+kb)^{\alpha-k}.$$

4. Show that if $f^{-1}(x)$ is the inverse of $f(x)$ and $f(0) = 0$, then assuming the necessary analyticity of the functions,

$$f^{-1}(x) = x \left(\frac{x}{f(x)} \right)_{x=0} + \frac{x^2}{2!} \left(\frac{d}{dx} \left(\frac{x^2}{f(x)} \right) \right)_{x=0} + \cdots.$$

5. Show that

$$\log(1+x) = \sum_{n=1}^{\infty} \binom{n\beta-1}{n-1} \frac{y^n}{n}$$

when $y = x(1+x)^{-\beta}$.

6. Suppose that $x^{m+1} + ax - b = 0$. Show that

$$x = \frac{b}{a} - \frac{b^{m+1}}{a^{m+2}} + \frac{2m+2}{2!} \frac{b^{2m+1}}{a^{2m+3}} - \frac{(3m+2)(3m+3)}{3!} \frac{b^{3m+1}}{a^{3m+4}} + \cdots.$$

Use this formula to find a solution of $x^5 + 4x + 2 = 0$ to four decimal places of accuracy. When $m = 0$, this series reduces to the geometric series. Write this sum as a hypergeometric series.

7. Use the Lagrange inversion formula to derive the generating function for Laguerre polynomials.

F

Series Solutions of Differential Equations

F.1 Ordinary Points

For readers not familiar with series solutions of differential equations, we give a few basic definitions so that the discussions of the hypergeometric equation in Chapter 2 and of the confluent and Bessel's equations in Chapter 4 are intelligible.

Consider the differential equation

$$a(x)\frac{d^2y}{dx^2} + b(x)\frac{dy}{dx} + c(x)y = 0 \quad (\text{F.1.1})$$

with $a(x)$, $b(x)$, and $c(x)$ analytic in the neighborhood of $x = x_0$. We take x to be a complex variable in this discussion. The simplest situation occurs when $a(x_0) \neq 0$. In this case, x_0 is called an ordinary point of the Equation (F.1.1).

It is not difficult to show that if x_0 is an ordinary point of (F.1.1), then (F.1.1) has a unique solution $f(x)$ analytic in a neighborhood of x_0 with prescribed values $f(x_0) = f_0$ and $f'(x_0) = f_1$. This implies that there are exactly two linearly independent solutions in a neighborhood of x_0 . To prove this result, it is convenient to divide (F.1.1) by $a(x)$ and rewrite the equation as

$$y'' = B(x)y' + C(x)y. \quad (\text{F.1.2})$$

Again for convenience, take $x_0 = 0$. Then

$$B(x) = \sum_{n=0}^{\infty} b_n x^n \quad \text{and} \quad C(x) = \sum_{n=0}^{\infty} c_n x^n.$$

Suppose that (F.1.2) has an analytic solution $f(x)$ with

$$f(x_0) = f_0, \quad f'(x_0) = f_1.$$

Then

$$f(x) = \sum_{n=0}^{\infty} f_n x^n. \quad (\text{F.1.3})$$

Substitute this series and the series for $B(x)$ and $C(x)$ in (F.1.2); equate the coefficients of x^n . The result is

$$(n+2)(n+1)f_{n+2} = \sum_{k=0}^n (n+1-k)f_{n+1-k}b_k + \sum_{k=0}^n f_{n-k}c_k, \quad (\text{F.1.4})$$

$n = 0, 1, 2, \dots$ This shows that f_0 and f_1 uniquely determine f_n for $n \geq 2$. It is now enough to prove that the series (F.1.3) has a positive radius of convergence. Since the series for $B(x)$ and $C(x)$ have a positive radius of convergence, there exist constants M and R such that

$$|b_n| \leq M/R^n \quad \text{and} \quad |c_n| \leq M/R^n.$$

We show by induction that there exist suitable positive numbers M_1 and r such that

$$|f_n| \leq M_1/r^n. \quad (\text{F.1.5})$$

This implies the needed result. Take M_1 such that $|f_1| \leq M_1$ and choose r such that $r < R$, $|f_1| \leq M_1/r$, and $M(r/2 + r^2) \leq 1$. Suppose $n \geq 2$ and assume (F.1.5) true up to $n-1$. By (F.1.4), the inequality (F.1.5) follows after a small calculation.

F.2 Singular Points

When $a(x_0) = 0$ and $b(x_0)$ and/or $c(x_0)$ is not zero, then $x = x_0$ is called a singular point of (F.1.1). Divide (F.1.1) by $a(x)$ and write it as

$$y'' + d(x)y' + e(x)y = 0. \quad (\text{F.2.1})$$

Consider first the simplest case where x_0 is a singular point and $a(x)$ has a simple zero at x_0 . Then $d(x)$ and $e(x)$ have at most simple poles. Take $x_0 = 0$. Then

$$d(x) = \sum_{n=-1}^{\infty} d_n x^n \quad \text{and} \quad e(x) = \sum_{n=-1}^{\infty} e_n x^n. \quad (\text{F.2.2})$$

Substitute the series (F.1.3) in (F.2.1) and equate coefficients as before. The reader may check that in this case we get only one solution, since the value of f_0 determines f_n for $n \geq 1$. Moreover, if d_{-1} is a nonnegative integer, then this method may fail to produce a solution.

Now consider the case where $d(x)$ has at most a simple pole and $e(x)$ at most a double pole. The simplest special case of (F.1.1) that leads to this is when

$$a(x) = a_2(x - x_0)^2, \quad b(x) = b_1(x - x_0), \quad c(x) = c_0. \quad (\text{F.2.3})$$

Two linearly independent solutions exist, at least one of the form $y = (x - x_0)^\mu$. To determine μ , substitute this expression for y in (F.1.1) to get

$$a_2\mu(\mu - 1) + b_1\mu + c_0 = 0. \quad (\text{F.2.4})$$

If this quadratic has two unequal roots, these roots give two independent solutions of (F.1.1). If the two roots equal μ_1 (say), then we have the solution $(x - x_0)^{\mu_1}$. To find the other independent solution, set $y = (x - x_0)^{\mu_1} w$. The differential equation for w has $\log(x - x_0)$ as a solution. Thus the second independent solution is $(x - x_0)^{\mu_1} \log(x - x_0)$.

F.3 Regular Singular Points

A point $x = x_0$ is called a regular singular point of (F.1.1) if

$$\lim_{x \rightarrow x_0} \frac{(x - x_0)b(x)}{a(x)} \quad \text{and} \quad \lim_{x \rightarrow x_0} \frac{(x - x_0)^2 c(x)}{a(x)}$$

both exist. If one of these limits does not exist, the singular point is irregular.

To see the difference between a regular and an irregular singular point, the easiest case to consider is the first-order analog of (F.1.1). A regular singular point occurs when

$$b(x)y' + c(x)y = 0 \quad (\text{F.3.1})$$

with $b(x)$ and $c(x)$ analytic in a neighborhood of $x = x_0$, and

$$\lim_{x \rightarrow x_0} (x - x_0)c(x)/b(x)$$

exists.

When $b(x) = (x - x_0)^2$ and $c(x) = c_0$, then

$$y(x) = Ae^{-c_0/(x-x_0)}$$

is the solution, and it has an essential singularity at $x = x_0$.

The simple case considered in the previous section, where $a(x)$, $b(x)$, $c(x)$ are as in (F.2.3), shows that, at a regular singular point, solutions may involve noninteger powers of $x - x_0$. Moreover, the second solution may have a logarithmic singularity.

Now suppose that x_0 is a regular singular point of (F.1.1) such that $a(x)$ has a double zero at x_0 . Then

$$a(x) = \sum_{n=2}^{\infty} a_n (x - x_0)^n, \quad b(x) = \sum_{n=1}^{\infty} b_n (x - x_0)^n, \quad \text{and} \quad c(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^n.$$

Suppose $y = (x - x_0)^\mu f(x)$ is a solution of (F.1.1). The equation for $f(x)$ can be seen to be

$$\begin{aligned} a(x)f'' + \left(2\mu \sum_{n=1}^{\infty} a_{n+1}(x - x_0)^n + b(x)\right)f' \\ + \left(c(x) + \mu \sum_{n=0}^{\infty} b_{n+1}(x - x_0)^n + \mu(\mu - 1) \sum_{n=0}^{\infty} a_{n+2}(x - x_0)^n\right)f = 0. \end{aligned} \quad (\text{F.3.2})$$

Now if μ is chosen so that

$$\mu(\mu - 1)a_2 + \mu b_1 + c_0 = 0, \quad (\text{F.3.3})$$

then (F.3.1) is an equation of the same type as the one considered at the beginning of Section F.2. In that case a solution of the form $f(x) = \sum_{n=0}^{\infty} f_n(x - x_0)^n$ is always possible provided $2\mu + b_1/a_2$ is not a nonnegative integer.

Note that the quadratic (F.3.2) has two solutions μ_1 and μ_2 . Thus it may be possible to obtain two solutions of the form $(x - x_0)^\mu f(x)$. It follows from (F.3.2) that

$$\begin{aligned} \mu_1 - \mu_2 &= 2\mu_1 - (\mu_1 + \mu_2) \\ &= 2\mu_1 - (1 - b_1/a_2) \\ &= 2\mu_1 + b_1/a_2 - 1. \end{aligned}$$

The remarks in the previous paragraph now imply that if the difference of the two roots of (F.3.2) is not an integer, then (F.1.1) has two independent solutions of the form $\sum_{n=0}^{\infty} f_n(x - x_0)^{n+\mu}$. However, if $\mu_1 - \mu_2 \geq 0$ is an integer, then a solution of this form with $\mu = \mu_1$ exists. The other solution may involve a logarithmic singularity.

Equation (F.3.2) is called the indicial equation. In practice it is obtained by substituting the series $\sum f_n(x - x_0)^{n+\mu}$ in (F.1.1) and equating the coefficients of the lowest power of $x - x_0$.

For the hypergeometric equation (2.3.5), there are three regular singular points 0, 1, and infinity. For the confluent equation (4.1.3) and the Bessel equation (4.5.1), 0 is a regular singular point, and infinity is an irregular singular point.

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